An exposition of smooth anisotropic Calderón uniqueness from full boundary data

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Abstract

We present the argument of OpenAI's September 2026 preprint on the smooth anisotropic Calderón problem and its implication for full boundary data. For a compact connected smooth manifold of dimension at least three, two smooth Riemannian metrics with the same boundary metric and the same single zero-frequency Dirichlet-to-Neumann map are related by a smooth diffeomorphism fixing the boundary pointwise. The proof is included: product continuation of Green functions, a uniform estimate on shrinking spheres, and continuation of the resulting local isometry. We also give the angular spectral estimates and the passage from density-valued patch measurements to the ordinary full boundary operator explicitly. The uniqueness argument and its priority belong to OpenAI; this article is an attributed exposition.

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1 Introduction

Boundary measurements for a harmonic equation record the response to an imposed boundary value. In the anisotropic Calderón problem the unknown is a Riemannian metric, and the response is its Dirichlet-to-Neumann map. A diffeomorphism fixing the boundary leaves this map unchanged. The uniqueness question is whether this coordinate freedom accounts for every ambiguity in smooth metrics when the dimension is at least three.

This article gives an exposition of the affirmative argument in OpenAI's September 24, 2026 preprint [5]. That preprint treats measurements supported in any nonempty boundary patch and, as a consequence, full boundary measurements. We include the patch proof, then deduce the formulation with one ordinary zero-frequency boundary map in Theorem 2.1. The manifold is compact and connected; its boundary need not be connected, and the manifold need not be orientable. All metrics are smooth and positive definite up to the boundary. The two metrics in the full boundary statement have the same tangential boundary metric, so their ordinary normal-derivative maps use the same boundary density. No further measurement is used.

The organizing question is how boundary agreement can be continued into the interior without analyticity. The Poisson embedding approach of Lassas, Liimatainen, and Salo [4] provides useful context: harmonic functions generated by sources encode points and local coordinates. Here we follow OpenAI's Green-function construction. Boundary symbol recovery first identifies every metric jet on the observed patch. A common cap then gives a region on which the two metrics and source kernels agree.

The main difficulty is to continue this agreement while a Green-function pole approaches a shrinking sphere. Qualitative unique continuation alone does not provide the needed uniform bound. OpenAI's argument continues the Green kernels on a product of the two manifolds and represents evaluation by a density on a moving sphere. An angular estimate bounds this density independently of the sphere's shrinking radius. Finite harmonic Hessian tests then recover the local metric, and an open-and-closed argument continues the resulting isometry throughout the manifold. We give the angular spectral mapping argument in Subsection 3.4 before it is used in that estimate.

The coverage of this exposition is the complete smooth uniqueness argument and its classical analytic inputs. The cap construction, product continuation, shrinking-sphere estimate, and globalization are due to OpenAI. The cited preprint is a source argument; this article does not assert independent human certification of it. The proof uses positive densities throughout, so local coordinate arguments do not impose a global orientation. The final section returns to the full ordinary Dirichlet-to-Neumann map and fixes every boundary point.

2 The full boundary problem and the patch theorem

Let M be a compact connected smooth manifold with nonempty smooth boundary, and let $n = \dim M \geq 3$. All metrics below are smooth and positive definite up to the boundary. Neither orientability of M nor connectedness of its boundary is assumed. We use positive densities and the convention $\Delta_g = \operatorname{div}_g \nabla_g$. For $f \in H^{1/2}(\partial M; \mathbb{C})$, let u_f be the unique weak solution of $\Delta_g u_f = 0$ with trace f. The ordinary Dirichlet-to-Neumann map is $\Lambda_g f = \partial_{\nu_g} u_f$, with outward unit normal. For smooth inputs this is a smooth boundary function; its weak extension uses the boundary density dS_g.

Theorem 2.1 (Full boundary uniqueness)**.** *For every such M and every pair g_1, g_2 inducing the same tangential metric on ∂M, equality of the single zero-frequency full Dirichlet-to-Neumann maps $\Lambda_{g_1} = \Lambda_{g_2}$ implies that there is a smooth diffeomorphism $F : M \to M$ with $F|_{\partial M} = \mathrm{Id}$ and $g_1 = F^* g_2$.*

The proof follows OpenAI's argument for measurements on one boundary patch [5, Theorem 1.1 and Corollary 1.2]. The patch theorem is proved in full below; it is not assumed as a

separate input. The final deduction records explicitly the passage from ordinary normal derivatives to density-valued measurements. The cap, product continuation, moving-sphere estimate, and global continuation are due to OpenAI. The exposition below retains that attribution.

For a nonempty relatively open $\Gamma \subset \partial M$, put

$$H^{1/2}_{\mathrm{co}}(\Gamma) = \{f \in H^{1/2}(\partial M) : \operatorname{supp} f \subset \Gamma\}, \qquad Q_g(f,h) = \int_M \langle du_f, d\overline{u_h}\rangle_g \, dV_g.$$

The supported space carries the inherited $H^{1/2}$ norm. It is not being defined as the closure of all supported smooth functions. Each individual support is compact in Γ. A cutoff equal to one near that support, followed by smooth approximation on the boundary, gives approximation within a fixed compact subset of Γ. The energy map is the distribution density on Γ defined by Q_g; for smooth inputs it is $\partial_{\nu_g} u_f \, dS_g$. Equality of these maps means equality of the sesquilinear forms for every pair in the supported space.

Theorem 2.2 (Patch uniqueness)**.** *Let M, n, g_1, g_2 be as above, without prescribing a common boundary metric. For every nonempty proper relatively open $\Gamma \subset \partial M$, equality $Q_{g_1}(f,h) = Q_{g_2}(f,h)$ for all $f, h \in H^{1/2}_{\mathrm{co}}(\Gamma)$ implies a smooth diffeomorphism $\Phi : M \to M$ fixing Γ pointwise and satisfying $g_2 = \Phi^* g_1$.*

All source-generated functions and the finite harmonic Hessian family will be selected before spatial dilation. The spatial dilation will then be fixed before any radius tends to zero. This order is needed because qualitative continuation away from the Green kernel's pole alone gives no uniform estimate as the separation vanishes.

3 Analytic preliminaries

We record the analytic inputs used in the proof, including the parameter estimates needed for moving balls. Sobolev spaces are defined in smooth charts by $\|v\|^2_{H^s} = \int \langle\xi\rangle^{2s} |\widehat{v}(\xi)|^2 \, d\xi$, using $\widehat{v} = (2\pi)^{-n/2} \int e^{-ix\cdot\xi} v(x) \, dx$. A finite partition of unity defines the norms on compact manifolds. Different smooth charts and positive smooth densities give equivalent norms; all such comparisons below are on fixed compact sets. The measure-theoretic inputs are monotone and dominated convergence, Tonelli and Fubini on sigma-finite spaces, and completeness of L^2; see [1, Theorems 3.11, 3.31, 5.28, 5.32 and 7.24]. All measures used here are positive smooth densities, Euclidean Lebesgue measure or counting measure, hence sigma-finite. Fubini is used with an absolute-integrability bound; nonnegative square integrals use Tonelli. Complex formulas follow by real and imaginary parts. Sobolev spaces are Hilbert completions, and local PDE regularity fixes their smooth representatives when point values are used.

3.1 Dirichlet solutions and uniform estimates

The Fourier inversion and Plancherel statements used here are [8, Propositions 3.1–3.2]; the corresponding Fourier series identity is given there in Section 1. For a half-space, the operator

$$Ef(x', s) = \mathcal{F}^{-1}_{\xi \to x'}(e^{-s\langle\xi\rangle} \widehat{f}(\xi)), \qquad s \geq 0,$$

followed by a fixed cutoff, is a bounded right inverse of the trace $H^1 \to H^{1/2}$. Indeed integration in s gives $\|Ef\|^2_{H^1} \leq C \int \langle\xi\rangle |\widehat{f}|^2$. Conversely the one-dimensional fundamental theorem of calculus gives

$$\langle\xi\rangle |\widehat{u}(\xi, 0)|^2 \leq \int_0^\infty \left(\langle\xi\rangle^2 |\widehat{u}|^2 + |\partial_s \widehat{u}|^2\right) ds$$

for smooth functions of compact support. Density and boundary charts give the trace and extension assertions on M. Convolution after local extension gives smooth approximation.

Zero extension across a flat boundary with zero trace has no distributional jump; conversely translations followed by convolution approximate such extensions by functions supported on the interior side. Applying cutoffs proves $\ker \mathrm{Tr} = H_0^1(M)$ and the gluing statement for H^1 functions with equal traces on a smooth interface.

For completeness the coercivity used here is also present in [7, Section 1, (1.16)]: for a fixed trace g there are $B > 0, C_g$ with $\frac{1}{2}\|du\|_2^2 \geq B\|u\|_2^2 - C_g$ on that trace class. With $g = 0$, replace u by au and let $a \to \infty$. Thus $\|u\|_2 \leq C\|du\|_2$ on $H_0^1(M)$. The form $a_g(u, v) = \int \langle du, d\bar{v}\rangle_g dV_g$ therefore defines an equivalent Hilbert norm. The Riesz representation theorem [9, Proposition 2.2] gives a unique zero-trace solution of $L_g u = F$ for every continuous functional F on H_0^1. Applying it to $-a_g(Ef, \cdot)$ gives the harmonic extension of f. Its energy is at most that of Ef, and hence Q_g is bounded on $H^{1/2} \times H^{1/2}$. Symmetry of a_g gives a self-adjoint Dirichlet inverse on smooth sources.

Here is the quantitative regularity fact used below. For a smooth uniformly positive metric on a fixed smooth compact domain and integer $q \geq 0$, its zero-Dirichlet solution obeys

$$\|u\|_{H^{q+2}} \leq C_q(\|L_g u\|_{H^q} + \|u\|_{H^1}). \tag{3.1}$$

The constants depend on finitely many coefficient bounds, ellipticity, and fixed boundary charts. One can derive this estimate directly as follows. In a flattened boundary chart the weak equation is in divergence form with symmetric coefficients a^{ij} and $a^{nn} > c > 0$. For a tangential difference quotient $D_{h,k}$, test the equation with $-D_{-h,k}(\eta^2 D_{h,k} u)$. Integration of difference quotients moves $D_{h,k}$ to $a^{ij}\partial_j u$; its principal term is

$$\int \eta^2 a^{ij}(x + he_k)\partial_j D_{h,k} u \, \partial_i D_{h,k}\bar{u}.$$

The coefficient difference and cutoff terms are bounded, by Cauchy–Schwarz and Young's inequality, by half that positive term plus $C(\|u\|_{H^1}^2 + \|L_g u\|_2^2)$. The source term uses $\|D_{-h,k}(\eta^2 D_{h,k} u)\|_2 \leq C\|\nabla(\eta^2 D_{h,k} u)\|_2$. Passing to $h \to 0$ controls every second derivative with at least one tangential index. The equation solves for $\partial_n^2 u$, so $u \in H^2$. For higher orders apply the same argument to each tangential derivative; its boundary trace is zero. Commutators contain derivatives of coefficients and lower derivatives of u. Recover derivatives with two or more normal indices successively from $a^{nn}\partial_n^2 u = -L_g u - \sum_{(i,j)\neq(n,n)} a^{ij}\partial_i\partial_j u$ plus the first-order terms. Induction first on total order and then on normal order proves (3.1). Interior patches use unrestricted difference quotients. A finite partition and slightly nested cutoffs give the global estimate. Approximation of the coefficients and source, or first applying the inequalities to difference quotients of the weak solution, justifies each step before the asserted derivatives exist. Nonzero smooth trace is subtracted by a smooth extension. In particular, smooth data give solutions smooth up to the boundary, in agreement with [7, Propositions 1.2–1.3].

The same local estimates with cutoffs give $\|u\|_{H^q(K)} \leq C_{q,K}\|u\|_{L^2(D)}$ for homogeneous solutions, $K \Subset D$, including scalar elliptic equations with bounded smooth lower-order terms. The initial H^1 estimate is obtained by testing with $\eta^2\bar{u}$; each further derivative follows from difference quotients. For distributional solutions local regularity instead follows from the elliptic parametrix described below.

Pull a smooth family of balls back to the fixed unit ball, including its limit at radius zero. For each q, (3.1) and coercivity give uniform bounded inverses $L(\theta)^{-1} : H^q \to H^{q+2} \cap H_0^1$ on compact parameter sets. Multiplying a metric Laplacian by a positive smooth function changes its source by that function's reciprocal, preserving these assertions. A perturbation in this operator norm is inverted by the Neumann series. Differentiation gives $\partial_\theta R = -R(\partial_\theta L)R$, and its iterates give smooth parameter dependence in every fixed Sobolev order. Sobolev embedding here follows directly from Fourier inversion and $\int \langle\xi\rangle^{-2s} < \infty$ for $s > n/2$. These estimates control smooth-kernel remainders by taking sufficiently large, fixed orders; the number of orders may depend on the requested seminorm, but no shrinking parameter enters that choice.

3.2 Symbols, Green kernels, and point distributions

Use classical scalar $\Psi^m = \Psi^m_{1,0}$ operators. The symbol product, adjoint, H^s boundedness, and elliptic parametrix are those of [6, Lecture I, Sections 2–4 and 8]. In particular, left symbols satisfy $b\#c \sim \sum_\alpha (1/i)^{|\alpha|}(\partial_\xi^\alpha b)(\partial_x^\alpha c)/\alpha!$; a scalar commutator lowers the summed order by one, and an operator of order m maps H^s boundedly to H^{s-m}. An elliptic operator of order two has a two-sided local parametrix of order -2 modulo smoothing. Its kernel is smooth off the diagonal. For that parametrix, [6, Lecture I, (1.23)–(1.24)] gives $|\partial_x^\alpha \partial_y^\zeta K(x,y)| \le C|x-y|^{2-n-|\alpha|-|\zeta|}$ when $n \ge 3$. Subtracting the true Dirichlet inverse leaves a smooth kernel on compact interior subsets: apply local regularity to the parametrix error, and to the transposed error in the second variable. Thus these bounds apply locally to the Dirichlet Green kernel. After dilation, each derivative has its matching power of the dilation, so compact subsets at fixed positive separation have uniform bounds. The smooth remainder is harmless since λ^{n-2} and its differentiated factors are bounded for $n \ge 3$. The weak inverse and its transpose supply the Green distribution; this construction supplies its off-diagonal smooth representative.

The boundary symbol factorization used below is the scalar case of [2, Theorem 1.1(i)–(ii) and Proposition 3.1]. The theorem is stated for an orientable manifold and all form degrees; we use its local scalar construction in half-balls and normal unit balls, which are orientable. Its recursion also applies after multiplication by a positive function: the harmonic solutions are unchanged, and any chosen transverse derivative trace is a smooth multiple of the unit normal trace plus a tangential derivative. Its dependence on parameters is obtained by the symbol recursion and the uniform inverse estimates above. No global differential-form conclusion is used. The authors' errata change the sign in a differential-form Green identity and correct an introductory reference, leaving this scalar construction unchanged.

A point-supported distribution is a finite sum of derivatives of a point mass [8, Proposition 4.5]. Locally this also follows without an additional representation theorem. If it has order k and all derivatives of a test function through k vanish at the point, multiplying that function by a cutoff at scale a gives C^k norm $O(a)$. The distribution's action tends to zero, and equals its action on the original function by point support. It therefore factors through the finite k-jet space, whose dual is spanned by derivatives of the point mass. The principal symbol calculation in the next section will use this fact for the transposed Dirichlet inverse.

3.3 Continuation and geometric coordinates

The open-set unique continuation input is [3, Theorem 8, Definitions 8.1–8.3] for scalar elliptic operators, applied in smooth coordinate balls. With positive quadratic principal symbol, at a nonzero normal covector the simultaneous complex conditions $p_\theta = \{p_\theta, \theta\} = 0$ have no solution: the imaginary part of the latter bracket is, up to sign, $2\tau|d\theta|_g^2$. Thus every smooth hypersurface is strongly pseudoconvex for the surface form of that theorem. One-sided zero sets propagate across coordinate sphere fronts, and a path with overlapping balls gives open-set continuation on a connected domain. Distributional solutions are first made smooth by the elliptic parametrix. The strict-weight Carleman estimate used for product continuation is [3, Definition 8.3 and Theorem 9(a)]; its specialization, normalization, and uniform margins are also derived in the proof below.

Local existence of geodesics and smooth parameter dependence follow by writing the geodesic equation as a first-order smooth system, using [10, Theorem 2.1, Propositions 6.1–6.2]. The inverse function theorem is [10, Theorem 3.1]. At zero velocity the exponential map has derivative the identity; these inputs give normal coordinate balls uniform on compact center sets. For a compact smooth boundary, the inward normal exponential has invertible differential at the zero section. Local injectivity and a finite cover give a uniform collar; otherwise two colliding sequences would limit to one boundary point and contradict local injectivity there. Differentiating the geodesic equation gives Jacobi fields and the $O(r^2)$ center-variation remainder used for the

moving traces. Metric compatibility shows $g(\partial_s, \partial_{z_a}) = 0$ along a normal collar: its derivative in s is $\frac{1}{2}\partial_{z_a} g(\partial_s, \partial_s) = 0$, and it is zero at $s = 0$. Hence $|ds|_g = 1$. Any curve reaching the boundary in the collar has length at least s, while the normal segment has length s; curves first leaving a fixed larger collar have a longer fixed lower bound when s is small. The collar parameter is consequently the boundary distance. This also proves the boundary-distance smoothness used in completing the final isometry. Green identities and metric normal transport use the coordinate divergence formula and integration by parts against positive densities; orientation is never needed.

3.4 Angular norms without fractional functional calculus

The shrinking-sphere estimate uses half-integer angular norms. We give the precise mapping statement needed there, including its uniformity. Let B be the outward radial Dirichlet-to-Neumann operator of the Euclidean unit ball in $\mathbb{R}^n$. Let Π_k project onto restrictions of harmonic homogeneous polynomials of degree k to S^{n-1}, with its probability measure. Then

$$B\Pi_k = k\Pi_k, \qquad \|v\|_s^2 = \sum_{k\geq 0} (1+k)^{2s} \|\Pi_k v\|_2^2.$$

Lemma 3.1 (Spectral angular mapping)**.** *For every integer $m \geq 0$ and real s, a scalar classical operator $A \in \Psi^m(S^{n-1})$ satisfies*

$$\|Av\|_{s-m} \leq C\|v\|_s. \tag{3.2}$$

The constant is bounded by finitely many symbol and smooth-kernel seminorms of A. Thus the bound is uniform for bounded smooth families on the fixed sphere, and applies componentwise to finitely many tensor components. Orders $m = 0, 1, 2, 3$ suffice below.

Proof. We first justify the complete decomposition used in the displayed norm. Restrictions of polynomials are dense in continuous functions on the sphere. Indeed extend a continuous function to a continuous compactly supported function F in an annulus of $\mathbb{R}^n$. Gaussian convolution converges uniformly to F on the sphere. On $|x| = 1$ its kernel is

$$e^{-|x-y|^2/a} = e^{-(1+|y|^2)/a} e^{2x\cdot y/a}.$$

For fixed $a > 0$, the exponential series converges uniformly on the sphere times the compact support of F. Integrating a finite partial sum gives a polynomial in x, approximating that convolution uniformly. Continuous functions are dense in L^2 by local convolution and cutoffs.

Every homogeneous polynomial p of degree k is a sum of $|x|^{2j}$ times harmonic homogeneous polynomials. To see this without an additional decomposition theorem, put

$$H = p + \sum_{j=1}^{\lfloor k/2 \rfloor} c_j |x|^{2j} \Delta^j p, \quad c_0 = 1, \quad c_j = -\frac{c_{j-1}}{2j(2k-2j+n-2)}.$$

The formula

$$\Delta(|x|^{2j} q) = 2j(2\deg q + 2j + n - 2)|x|^{2j-2} q + |x|^{2j} \Delta q$$

shows that $\Delta H = 0$. The difference $p - H$ is divisible by $|x|^2$; induction on degree completes the decomposition. In polar coordinates, $-\Delta_{S^{n-1}} H = k(k+n-2)H$. Integration by parts gives orthogonality of distinct degrees. These finite-dimensional spaces therefore give a complete orthogonal decomposition. The harmonic extension is $r^k H(\omega)$, proving $B\Pi_k = k\Pi_k$, including $k = 0$.

By the local scalar boundary construction already recorded, B is a classical elliptic operator of order one with principal symbol $|\xi|$. For an integer $m \geq 0$, a parametrix for $(1+B)^m$ and the ordinary chart-Sobolev mapping bounds give

$$\|u\|_{H^m} \leq C_m(\|(1+B)^m u\|_2 + \|u\|_2) \leq 2C_m \|(1+B)^m u\|_2.$$

Here the smoothing remainder is bounded $L^2 \to H^m$, and the last inequality uses $1 + k \geq 1$. This step uses only integer powers.

Put $P_j = \sum_{2^j \leq 1+k<2^{j+1}} \Pi_k$, $j \geq 0$. Since a commutator with the scalar order-one operator B preserves the order m, all $\operatorname{ad}_B^N(A)$ belong to Ψ^m. For each fixed integer $N \geq 0$, the preceding integer estimate implies

$$\|\operatorname{ad}_B^N(A) P_l\|_{2\to 2} \leq C_N 2^{lm}. \tag{3.3}$$

Each C_N uses only finitely many seminorms of A. For $|j - l| \geq 2$ the two finite-dimensional spectra of $B_j = B|_{P_j}$ and $B_l = B|_{P_l}$ have distance at least $c \max(2^j, 2^l)$. If $j > l$, set $X = P_j A P_l$ and $Y = P_j [B, A] P_l$. The finite-dimensional identity $B_j X - X B_l = Y$ gives

$$X = \int_0^\infty e^{-tB_j} Y e^{tB_l}\, dt, \qquad \|X\| \leq \frac{\|Y\|}{\operatorname{dist}(\operatorname{spec} B_j, \operatorname{spec} B_l)}.$$

It follows by differentiating $e^{-tB_j} X e^{tB_l}$; the exponential tends to zero because of the spectral gap. For $j < l$, use the opposite sign and the exponentials e^{tB_j} and e^{-tB_l}. Iteration and (3.3) yield

$$\|P_j A P_l\|_{2\to 2} \leq C_N 2^{lm} \max(2^j, 2^l)^{-N}, \quad |j - l| \geq 2. \tag{3.4}$$

For $|j - l| \leq 1$, the bound $C 2^{lm}$ suffices.

On each band the norm $\|\cdot\|_s$ is equivalent, with factors at most $2^{|s|}$, to $2^{js}\|\cdot\|_2$. Multiplying the block bound by $2^{j(s-m)-ls}$ leaves, in the far case,

$$C_N 2^{(j-l)(s-m)} 2^{-N \max(j,l)} \leq C_N 2^{-(N-|s-m|)|j-l|}.$$

Choose $N > |s - m| + 2$. The resulting nonnegative matrix b_{jl} has uniformly bounded row sums and column sums, including its three near-diagonal entries. For finitely supported numbers a_l,

$$\sum_j \left(\sum_l b_{jl} |a_l| \right)^2 \leq \left(\sup_j \sum_l b_{jl} \right) \left(\sup_l \sum_j b_{jl} \right) \sum_l |a_l|^2,$$

by Cauchy–Schwarz in each row. This proves (3.2) first for finite harmonic sums, then by completion. For a smooth input, integration by parts with every integer power of $-\Delta_{S^{n-1}}$ shows rapid weighted square summability of its harmonic components. The finite sums therefore converge in every integer chart-Sobolev norm by the integer estimate above. The completed operator agrees with the original A on smooth inputs. In particular no exchange involving an unbounded operator precedes the estimate. The proof is uniform in the stated seminorms. □

Spectral Cauchy–Schwarz gives the dual pairing $|\langle u, v\rangle| \leq \|u\|_{-s}\|v\|_s$. Consequently angular operators of orders one, two and three have the form bounds

$$|\langle A_1 u, v\rangle| \leq C\|u\|_{1/2}\|v\|_{1/2}, \qquad |\langle A_2 u, v\rangle| \leq C\|u\|_{3/2}\|v\|_{1/2}, \qquad |\langle A_3 u, v\rangle| \leq C\|u\|_{3/2}\|v\|_{3/2}.$$

Applying the same bounds to $A_2^* \in \Psi^2$ gives the second form bound with the two input orders reversed. An orthonormal identification of a tangent sphere with the fixed Euclidean sphere is unitary and conjugates B to this same operator. Thus the bounds hold pointwise in the center with the uniform symbol constants specified in the moving-sphere argument. They introduce no additional center derivatives of the small tensor.

4 Boundary data and a common interior region

We first convert the measurements into data that can be continued in the interior. Equality of the energy maps determines the full boundary jets on the measured patch. These jets allow us

to attach the same exterior cap to both metrics. Sources in that cap then supply common Green data and the finite families of harmonic functions needed in the local proof.

We use $\Delta_g = \operatorname{div}_g \nabla_g$ and positive Riemannian densities throughout. In particular, no orientation is chosen. The Dirichlet realization of $L_g = -\Delta_g$ on a compact connected smooth manifold with nonempty boundary is invertible: its quadratic form on H_0^1 is coercive. Smooth sources and boundary values give solutions smooth up to the boundary. We use interior elliptic regularity and unique continuation for smooth scalar second-order elliptic operators with real positive principal part, including a metric Laplacian multiplied by a positive smooth function. Here unique continuation means that an existing solution which vanishes on a nonempty open subset of a connected domain vanishes throughout that domain; see [3, Theorem 8]. For a positive scalar elliptic principal symbol, the local noncharacteristic-surface condition applies at every smooth hypersurface; propagation through overlapping coordinate balls gives this open-set form of continuation.

We will also use its boundary version. If a smooth harmonic function has both zero value and zero normal derivative on an open boundary patch, extend the metric smoothly across that patch and the function by zero. Green's formula shows that the extended function solves the equation distributionally: neither a value jump nor a flux jump remains. Elliptic regularity and interior unique continuation give vanishing in a neighborhood of the patch, and then in the connected interior.

4.1 The density and the boundary jets

The distinction between the measured energy operator and the ordinary unit-normal DN operator matters in the first step. The latter sends f to $\partial_{\nu_g} u_f$; the former sends it to the density $\partial_{\nu_g} u_f \, dS_g$.

Lemma 4.1 (Boundary jets)**.** *Let g_1, g_2 be smooth metrics on a compact connected smooth n-manifold M with nonempty boundary, where $n \geq 3$. Suppose their energy maps agree for inputs and observations in a nonempty relatively open set $\Gamma \subset \partial M$. In boundary-normal coordinates for the respective metrics, based on the same coordinates on Γ, their complete boundary Taylor jets agree.*

Proof. Fix a boundary chart compactly contained in Γ, write $m = n-1 \geq 2$, and let h denote the tangential metric matrix. Relative to the coordinate density, the energy operator has principal symbol

$$e_1(z, \xi) = \sqrt{\det h(z)} \sqrt{h^{ab}(z) \xi_a \xi_b}.$$

Thus its square determines the positive matrix

$$C = (\det h) h^{-1}, \qquad \det C = (\det h)^{m-1}, \qquad h = (\det C)^{1/(m-1)} C^{-1}.$$

The two induced metrics, and hence their boundary densities, are therefore equal. Dividing the common density-valued operator by this known density gives equality of the ordinary DN operators locally. Cutoffs supported in Γ and equal to one near a chosen point give equality of their complete symbols there.

We give the scalar symbol recursion to make its smooth and local content explicit, using Joshi and Lionheart [2, Theorem 1.1(ii) and Proposition 3.1]. In inward normal coordinates $s \geq 0$, write

$$g = ds^2 + h(s, z), \qquad a = \frac{1}{2} \operatorname{tr}(h^{-1} \partial_s h), \qquad L_g = -\partial_s^2 - a \partial_s + P,$$

where $P = -\Delta_{h(s)}$ is tangential. Smooth boundary factorization gives a tangential operator $B(s)$ of order one, with positive principal symbol, such that

$$L_g = (-\partial_s + B - a)(\partial_s + B) \pmod{\Psi^{-\infty}}, \qquad B^2 - aB - \partial_s B = P \pmod{\Psi^{-\infty}}. \tag{4.1}$$

Here the remainders are tangentially smoothing and depend smoothly on s. The local Poisson parametrix identifies $B(0)$, modulo smoothing, with the outward DN operator: the decaying branch satisfies $(\partial_s + B)u = 0$ and $\partial_\nu = -\partial_s$. This is a local factorization for smooth coefficients; it makes no analyticity assumption.

For precision, use left symbols and $D_z = -i\partial_z$, so

$$b \# c \sim \sum_\alpha \frac{(1/i)^{|\alpha|}}{\alpha!} (\partial_\xi^\alpha b)(\partial_z^\alpha c).$$

Write $b \sim b_1 + b_0 + b_{-1} + \cdots$ for the symbol of B and $p = p_2 + p_1$ for that of P. Then

$$b_1 = \rho = (h^{ab}\xi_a\xi_b)^{1/2}, \qquad b\#b - ab - \partial_s b = p.$$

The equation of degree one is

$$2\rho b_0 + \frac{1}{i}\partial_\xi \rho \cdot \partial_z \rho - a\rho - \partial_s \rho = p_1.$$

The terms p_1 and the tangential composition term involve only h and its tangential derivatives. Since

$$\partial_s \rho = -\frac{(\partial_s h)(\xi^\sharp, \xi^\sharp)}{2\rho},$$

the part of b_0 containing a normal derivative is

$$\frac{1}{4}\left(\operatorname{tr}_h \partial_s h - \frac{(\partial_s h)(\xi^\sharp, \xi^\sharp)}{\rho^2}\right).$$

At each subsequent degree the new coefficient is obtained from

$$2\rho b_{1-k} = \partial_s b_{2-k} + R_k, \qquad k \geq 2,$$

where R_k uses normal derivatives of h only through order $k-1$ and finitely many tangential derivatives. Indeed all other terms in the symbol product use earlier coefficients, and tangential differentiation does not increase normal derivative order. Writing $A_k = \partial_s^k h$, induction gives the highest-jet term

$$b_{1-k} = \frac{1}{4(2\rho)^{k-1}}\left(\operatorname{tr}_h A_k - \frac{A_k(\xi^\sharp, \xi^\sharp)}{\rho^2}\right) + \text{terms involving lower normal jets.} \tag{4.2}$$

Differentiating ρ, the raising of indices, or the coefficient in the preceding induction step contributes only lower-jet terms.

Suppose the two normal jets agree through order $k-1$ as smooth functions on the boundary chart. Their tangential derivatives also agree. Equality of the symbols and Equation (4.2) give, for $A = \partial_s^k (h_1 - h_2)|_{s=0}$,

$$\operatorname{tr}_h A - A(v, v) = 0 \qquad (|v|_h = 1).$$

Polarization yields $A = (\operatorname{tr}_h A)h$. Taking the trace gives $(m-1)\operatorname{tr}_h A = 0$, so $A = 0$. Starting from the already recovered boundary metric, induction proves all normal and mixed jet equalities. The normal metric components are $g_{ss} = 1$ and $g_{sa} = 0$, so these are the full metric jets.

Every step is local to a half-coordinate ball and uses densities. Although the differential-form formulation in [2] assumes orientability, this scalar coordinate proof requires none. Other boundary components affect only the local smoothing remainder. Finally, equality of smooth jets asserts no equality in an interior collar. □

4.2 Attaching a common cap

The jets provide a common region on the exterior side of a smaller patch. The following energy argument transfers the measured data to that region without making assumptions on the inaccessible boundary.

Proposition 4.2 (Common cap and Green data)**.** *Under the hypotheses of Lemma 4.1, there are compact connected smooth extensions (M_j^e, g_j^e) of (M, g_j) and an identified connected open exterior cap E on which the extended metrics agree. The cap is attached through a nonempty open set $P_0 \Subset \Gamma$. The extended zero-Dirichlet Green kernels satisfy*

$$G_1(x, y) = G_2(x, y), \qquad x, y \in E, \quad x \neq y.$$

For every $F \in C_c^\infty(E)$, the two extended zero-Dirichlet solutions with source F agree on E.

Proof. Choose a boundary coordinate patch $P \Subset \Gamma$ and a nonnegative smooth bump b with $\operatorname{supp} b \Subset P$ and connected, nonempty positivity set $P_0 = \{b > 0\}$. Use the separate inward normal collars of Lemma 4.1, and replace the boundary $s = 0$ by the graph $s = -b(z)$. Take b sufficiently small that this graph lies in a collar. It gives smooth compact connected extended manifolds. Their common open added part is

$$E = \{(z, s) : -b(z) < s < 0\}.$$

Extend the first metric smoothly to negative s and use the same exterior metric for the second. A smooth extension of its coordinate coefficients exists in a collar and remains positive definite after shortening the collar. Their jets match on an open neighborhood of $\operatorname{supp} b$. Pasting across the entire hypersurface $s = 0$ in that neighborhood is therefore smooth, to every order. Restricting this pasting to $s \geq -b(z)$ proves smoothness also at the edges of the bump. We henceforth omit the superscript on the extended metrics.

For the following comparison, identify the two extensions by the identity on the original M and by the common collar coordinates on E. This identification and its inverse are smooth up to each side of the seam, with bounded derivatives, and agree on the seam. In collar charts they are therefore bi-Lipschitz. In particular they identify the spaces H_0^1 on the two extensions.

Fix a real $F \in C_c^\infty(E)$ and let w_j minimize

$$\mathcal{J}_j(w) = \frac{1}{2} \int_{M_j^e} |dw|_{g_j}^2 \, dV_{g_j} - \int_E F w \, dV_{g_j} \quad \text{on } H_0^1(M_j^e).$$

The minimizers are smooth up to the extended boundary and solve $-\Delta_{g_j} w_j = F$. Their traces h_j on the original boundary are smooth and supported in $\operatorname{supp} b \Subset \Gamma$: wherever $b = 0$, that boundary is also the new zero-data boundary. Transport w_1 by this identification and replace it inside the original manifold by the g_2-harmonic extension of h_1, leaving it unchanged on E. The difference between the replacement and the transported w_1 is the zero extension of a function in $H_0^1(M)$; thus the replacement is an admissible competitor in $H_0^1(M_2^e)$. This argument does not require the narrowing cap to have a regular boundary at its edges.

The interior energies before and after replacement are the equal quadratic energies of h_1 for the two measured maps. The cap metric and source pairing are common. Hence $\min \mathcal{J}_2 \leq \min \mathcal{J}_1$. Reversing the roles gives equality. Strict convexity of the Dirichlet form makes the replacement equal to w_2, proving $w_1 = w_2$ on E.

Normalize the symmetric Dirichlet Green kernels by

$$-\Delta_{g_j, x} G_j(x, y) = \delta_y^{g_j}(x), \qquad G_j(\,\cdot\,, y)|_{\partial M_j^e} = 0, \qquad w_j(x) = \int_{M_j^e} G_j(x, y) F(y) \, dV_{g_j}(y), \quad (4.3)$$

where the point mass has unit mass relative to metric volume. The source densities are identical on E. Varying F gives equality of the distribution kernels on $E \times E$, and therefore pointwise equality off the diagonal. □

Fix a nonempty open source set $U_0 \Subset E$. Figure 1 shows the two boundaries and a possible choice of this set.

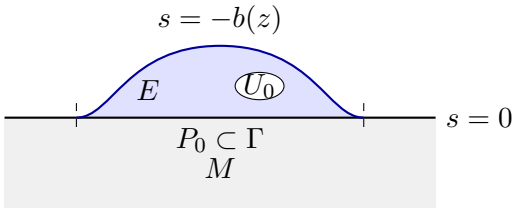


Figure 1: A local schematic cross-section of the common exterior cap. The boundary moves only over a compact subset of the measured patch. The oval is a nonempty open source set U_0 with closure inside E. The proof uses the smooth graph, including its flat bump edges.

4.3 Harmonic tests from sources in the cap

For $f \in C_c^\infty(U_0)$, let u_f^j be the zero-Dirichlet solution of $L_{g_j} u_f^j = f$ on M_j^e. Define its evaluation at a point by

$$I_j(p)(f) = u_f^j(p). \tag{4.4}$$

Equality of evaluations means equality for every such source f. Proposition 4.2 gives $I_1 = I_2$ on E. The functions u_f^j are harmonic away from $\overline{U_0}$; we only use their harmonic jets there.

This use of harmonic evaluations follows the Poisson and source embedding approaches of Lassas et al. [4, Sections 2 and 6.1]. The restricted-source approximation and separation results appear in [4, Appendix A, Propositions A.6–A.7]. Unique continuation forces the inverse of an annihilating point distribution to have point support, and differential order then determines that distribution. We give the full argument for the required second jets and positive densities.

Lemma 4.3 (Separation and finite harmonic jets)**.** *Let (N, g) be a compact connected smooth manifold with nonempty boundary and dimension $n \geq 2$. Let $U \Subset N^\circ$ be nonempty and open, and let $R = L_{g,D}^{-1}$. Then:*

(i) *The functionals $I(p) : f \mapsto Rf(p)$, $f \in C_c^\infty(U)$, separate distinct interior points. They are nonzero in the interior and zero on the boundary. For each fixed f, they depend continuously on $p \in N$.*

(ii) *If $p \in N^\circ \setminus \overline{U}$, the second jets of Rf at p fill exactly the hyperplane defined by $L_g u(p) = 0$. In particular they provide harmonic coordinates near p and any finite spanning family of harmonic Hessians in those coordinates.*

Proof. Continuity and the zero boundary values follow from smooth Dirichlet regularity. Non-vanishing follows by the same point-support argument used below: if $I(p) = 0$, then $w = R\delta_p$ vanishes on U and $L_g w = \delta_p$. Elliptic regularity and unique continuation on the punctured connected interior force w to be point-supported. A nonzero point-supported distribution has elliptic image of order at least two, and cannot have the order-zero image δ_p. If $p \in U$, the equation on U already contradicts $w|_U = 0$.

We prove the jet assertion by finite-dimensional duality. Let T be a linear functional on second jets at p annihilating all Rf, represented as a distribution of order at most two supported at p. Define $w = RT$ by transposition with the metric density. The Dirichlet inverse is self-adjoint, so

$$\langle w, f \rangle = \langle T, Rf \rangle = 0, \qquad f \in C_c^\infty(U).$$

Thus $w = 0$ on U and $L_g w = T$. Away from p it is smooth and harmonic by elliptic regularity. The punctured interior is connected: a path meeting p can be diverted in a small punctured

coordinate ball, whose spheres are connected in dimension at least two. Unique continuation consequently gives $\operatorname{supp} w \subset \{p\}$.

A distribution supported at one point is a finite sum

$$w = \sum_{|\alpha| \le k} c_\alpha \partial^\alpha \delta_p.$$

If $w \neq 0$ and k is its highest derivative order, then $L_g w$ has exact order $k+2$. Indeed its highest coefficient polynomial is the product of the nonzero degree-k coefficient polynomial and the nonzero elliptic quadratic principal symbol. Such a product cannot vanish. Since T has order at most two, $w = c\delta_p$. Hence the annihilator consists precisely of multiples of $u \mapsto L_g u(p)$. That functional does annihilate the source solutions because $p \notin \overline{U}$. Finite-dimensional duality proves that their jet range is the whole stated hyperplane.

To prove separation, suppose $I(p) = I(q)$ for distinct interior points. Apply the same transposition argument to $T = \delta_p - \delta_q$. The solution RT vanishes on U, and UCP on the connected interior with these two points removed shows that it is supported at those points. The argument still applies if a point lies in U, since the initial vanishing on U is distributional. At each support point a nonzero point-supported distribution has an elliptic image of order at least two. It cannot have the order-zero image $\delta_p - \delta_q$. This contradiction proves separation.

Finally the equation constraint allows arbitrary first derivatives, by adjusting the Hessian. Choose n source solutions with independent differentials; their tuple is a harmonic coordinate chart. In this chart the first-order coefficients of Δ_g vanish, so the ordinary coordinate Hessians have precisely the constraint $g^{ab}\partial_{ab} u = 0$. The jet conclusion supplies a basis for that hyperplane, with zero first derivatives if desired. A chosen finite basis remains spanning, with a positive Gram bound, after shrinking the chart. □

Remark 4.4. At any point outside $\overline{U_0}$ we may use finitely many pairs (u_f^1, u_f^2) to supply the harmonic coordinates and Hessians in the local argument. Once chosen, these functions remain fixed while the spatial scales shrink. The constant pair $(1, 1)$ is adjoined separately; it is not a zero-boundary source solution. All these statements are scalar and use positive densities, so they retain nonorientable manifolds and arbitrary boundary components.

5 Local continuation and product solutions

5.1 The local metric extension theorem

The common cap gives an initial isometry. To enlarge it, the global argument will use source-generated harmonic coordinates to identify neighborhoods of two interior points. In these coordinates we must continue equality of the metrics across a point where equality is already known on one side. The following theorem is the local input.

Theorem 5.1 (Local metric extension)**.** *Let $n \ge 3$. Let (N_1, g_1) and (N_2, g_2) be smooth compact connected Riemannian manifolds with nonempty smooth boundary, with Dirichlet Green functions G_1, G_2. Identify two interior harmonic coordinate neighborhoods with one neighborhood $\Omega \subset \mathbb{R}^n$ of 0. Suppose an open $W \subset \Omega$ contains, near 0, one side of a smooth hypersurface through 0, and suppose*

$$g_1 = g_2 \text{ on } W, \qquad G_1(x,y) = G_2(x,y) \quad (x, y \in W,\ x \neq y).$$

Suppose there are finitely many smooth pairs u_j^a harmonic for g_j on Ω, agreeing on W, whose first-side coordinate Hessians at 0 span

$$\{H \in \operatorname{Sym}^2(\mathbb{R}^n) : g_1^{ij}(0) H_{ij} = 0\}.$$

Then the two coordinate metrics agree on a neighborhood of 0.

We prove this theorem in Section 8. Its proof first constructs a mixed Green kernel that is harmonic for the first metric in x and for the second metric in y: initially it is $G_1(x, y)$ when $y \in W$ and $G_2(x, y)$ when $x \in W$. For each prescribed compact region at positive separation from $x = y$, a sufficiently small spatial dilation allows the continuation constructed in this section to reach that region. Section 6 then continues it to a family of small spheres centered at y, under a temporary bound that controls a normalized difference of the inverse metrics, together with finitely many derivatives, by the current radius. Its flux transfers first-metric harmonic functions to second-metric harmonic functions. In particular, the resulting functional sends u_1^a to u_2^a and reproduces the constant and coordinate functions.

The remaining task is uniform control as the spheres shrink. Section 7 estimates the density representing the flux functional. Under the same temporary metric bound, this estimate and exact reproduction of affine functions give an L^2 bound for each paired harmonic difference divided by the squared radius. Section 8 combines it with the spanning harmonic Hessians to improve the temporary bound strictly. This improvement allows the shrinking parameter to tend to zero with the spatial dilation fixed, forcing the two metrics to be conformal near the contact point. The common harmonic coordinates remove the remaining conformal factor. The geometric continuation in this section provides existence away from the diagonal; its estimates need not remain bounded as the separation tends to zero.

5.2 Product solutions and extension from a cross

We begin with a statement for general scalar elliptic operators. Let L_i be a smooth scalar second-order elliptic operator in a coordinate domain, with real positive definite principal matrix g_i^{-1}. Lower-order terms are allowed. A *product solution* is a function $F(x, y)$ satisfying

$$L_1^x F = 0, \qquad L_2^y F = 0.$$

Such a function has ordinary unique continuation, since $L_1^x + L_2^y$ is elliptic on the product. We shall construct extensions explicitly; unique continuation will identify extensions already constructed.

The starting data consist of two compatible product sets, with one variable restricted to a smaller observed set in each piece. A common basis for a solution space and its restriction to the smaller set will construct the extension. We give the variable-coefficient singular-function argument, including its quantitative dependence.

Lemma 5.2 (Extension from a cross)**.** *Let D_i be bounded connected coordinate domains and $S_i \Subset D_i$ nonempty open sets. Suppose a product solution F is given consistently on*

$$(D_1 \times S_2) \cup (S_1 \times D_2),$$

with the sum of its two L^2 norms at most M. Let $O_i \Subset D_i$ be open sets such that every L_i-solution w on D_i satisfies

$$\|w\|_{L^2(O_i)} \le C \|w\|_{L^2(S_i)}^{\gamma_i} \|w\|_{L^2(D_i)}^{1-\gamma_i}, \qquad 0 < \gamma_i < 1, \qquad \gamma_1 + \gamma_2 > 1. \tag{5.1}$$

Then F extends as a product solution to $O_1 \times O_2$, agreeing with both given pieces on their intersections. On every compact subset of this product its C^m norm is at most $C_m M$. The constants are uniform when the geometric margins, ellipticity, coefficient bounds, interpolation constants and positive exponent margin are uniform. Each specified output order uses only finitely many coefficient bounds.

Proof. Let $\mathcal{H}_1(D_1)$ be the closed subspace of $L^2(D_1)$ consisting of distributional L_1-solutions. Restriction $R : \mathcal{H}_1(D_1) \to L^2(S_1)$ is compact and injective. To prove compactness, multiply each solution by a cutoff equal to one near $\overline{S_1}$ and compactly supported in D_1. Interior estimates

bound its H^q norm by its $L^2(D_1)$ norm. Fourier truncation in a containing cube gives a finite-rank approximation to R of rank $O(A^n)$ and operator error $O(A^{-q})$. Hence R is compact; injectivity follows by unique continuation. The spectral theorem for R^*R gives an orthonormal basis (w_j) of $\mathcal{H}_1(D_1)$ such that

$$(w_i, w_j)_{L^2(S_1)} = \mu_j^2 \delta_{ij}, \qquad 0 < \mu_j \le 1. \tag{5.2}$$

For every N,

$$\mu_j \le C_N (1+j)^{-N}. \tag{5.3}$$

Indeed let K be the rank-r approximation just constructed. Its restriction to the span of the first $r+1$ eigenvectors has a unit kernel vector v, for which $\|(R-K)v\| = \|Rv\| \ge \mu_{r+1}$. Thus $\mu_{r+1} \le \|R-K\|$. The Fourier bound with arbitrary integer q yields $\mu_j \le C_q j^{-q/n}$ and proves the assertion, with the stated uniformity. If the space is finite dimensional, the corresponding finite sum needs no decay argument.

For $y \in S_2$ expand $F(\cdot, y)$ on D_1:

$$F(x,y) = \sum_j w_j(x) b_j(y), \qquad \|b_j\|_{L^2(S_2)} \le M.$$

Equation (5.2) continues these coefficients to D_2 by the formula

$$b_j(y) = \mu_j^{-2} \int_{S_1} F(x,y) \overline{w_j(x)}\, \mathrm{d}x, \qquad \|b_j\|_{L^2(D_2)} \le M \mu_j^{-1}. \tag{5.4}$$

The integral solves $L_2 b_j = 0$ and agrees with the original coefficient on S_2. Applying (5.1) gives

$$\|w_j\|_{L^2(O_1)} \le C \mu_j^{\gamma_1}, \qquad \|b_j\|_{L^2(O_2)} \le C M \mu_j^{\gamma_2 - 1}.$$

Consequently the series converges absolutely in $L^2(O_1 \times O_2)$: its summands are bounded by $CM\mu_j^{\gamma_1+\gamma_2-1}$. Choose N in (5.3) so that $N(\gamma_1+\gamma_2-1) > 1$. The limit solves both equations in distributions. Interior estimates for their elliptic sum give the asserted smoothness and bounds on smaller sets.

We verify agreement with the second given piece; no assertion that an arbitrary local solution extends to D_1 is needed. Let z be orthogonal to the closure of $R\mathcal{H}_1(D_1)$ in $L^2(S_1)$. The function

$$y \longmapsto \int_{S_1} F(x,y) \overline{z(x)}\, \mathrm{d}x$$

solves the second equation and vanishes on S_2, so it vanishes on connected D_2. Thus each second-piece slice belongs to that closed range. Its expansion in the orthonormal basis w_j/μ_j is complete, and its coefficients are $\mu_j b_j(y)$ by (5.4). It follows that the series is F on $S_1 \times O_2$; the series converges there also by the positive exponent γ_2. On $O_1 \times S_2$ it is the original expansion, with positive exponent γ_1. This proves both agreements. Only upper bounds for the singular values were used, so uniformity does not require continuous choices of singular bases or positive lower bounds for their singular values. □

Figure 2 summarizes the sets and the two estimates in Lemma 5.2.

5.3 A geometric criterion for local extension

We next build the interpolation estimates needed in Lemma 5.2. For a metric g, call a smooth real function θ a *strict weight* at a point where $\mathrm{d}\theta \neq 0$ if

$$\mathrm{Hess}_g\,\theta(e,e) + \mathrm{Hess}_g\,\theta(v,v) > 0 \quad \text{for all } v \perp e,\ |v|_g = 1, \qquad e = \frac{\nabla^g \theta}{|\nabla^g \theta|_g}. \tag{5.5}$$

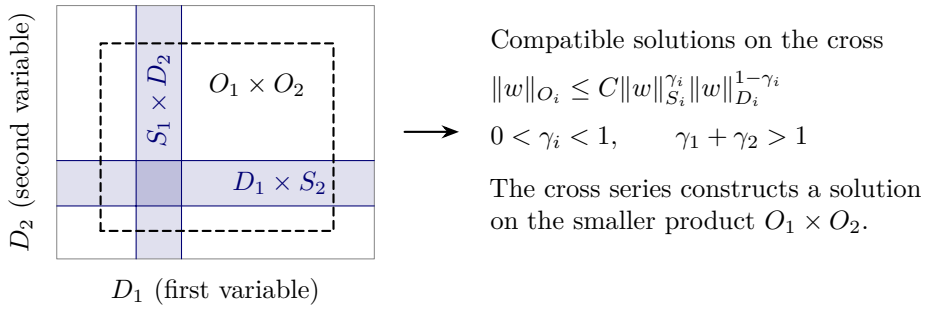


Figure 2: Compatible product solutions are given on the shaded cross. The L^2 interpolation estimates for single-variable solutions, with $\gamma_1 + \gamma_2 > 1$, make the singular-function series converge on $O_1 \times O_2$. The drawing is schematic: each factor is a multidimensional domain, and containment of S_i in O_i is not required.

The strict scalar Carleman estimate gives, after shrinking the coordinate neighborhood,

$$\tau^3\|e^{\tau\theta}w\|_{L^2}^2 + \tau\|\nabla(e^{\tau\theta}w)\|_{L^2}^2 \le C\|e^{\tau\theta}Lw\|_{L^2}^2, \qquad \tau \ge \tau_0, \tag{5.6}$$

for compactly supported w. This is the elliptic strict-weight estimate of the classical Carleman theory; see Koch and Tataru [3, Definition 8.3 and Theorem 9(a)]. Here the hypotheses can also be checked directly. For $L = -\Delta_g$, write $v = e^{\tau\theta}w$ and

$$e^{\tau\theta}Le^{-\tau\theta} = A_\tau + B_\tau, \qquad A_\tau = -\Delta_g - \tau^2|\mathrm{d}\theta|_g^2, \quad B_\tau = \tau(2\nabla^g\theta\cdot\nabla + \Delta_g\theta).$$

Here A_τ is self-adjoint and B_τ is skew-adjoint for metric volume. Integration by parts gives

$$\|(A_\tau + B_\tau)v\|^2 = \|A_\tau v\|^2 + \|B_\tau v\|^2 + ([A_\tau, B_\tau]v, v),$$
$$([A_\tau, B_\tau]v, v) = 4\tau\int \mathrm{Hess}_g\,\theta(\nabla v, \nabla\overline{v}) + 4\tau^3\int \mathrm{Hess}_g\,\theta(\nabla\theta,\nabla\theta)|v|^2 - \tau\int(\Delta_g^2\theta)|v|^2.$$

Choose a real constant m strictly between the negative of the least unit tangential Hessian value and the unit normal Hessian value. After shrinking the patch, these inequalities have a positive margin. Add $4m\tau(A_\tau v, v)$ to the commutator form. Its gradient coefficient becomes $4\tau(\mathrm{Hess}_g\,\theta + mg)$, and its leading zero-order coefficient is

$$4\tau^3|\mathrm{d}\theta|_g^2(\,\mathrm{Hess}_g\,\theta(e,e) - m) > c\tau^3.$$

The tangential gradient form is positive. Its mixed and possibly negative normal terms cost at most $C\tau\|\nabla_e v\|^2$, and

$$\|\nabla_e v\|^2 \le C\tau^{-2}\|B_\tau v\|^2 + C\|v\|^2.$$

Finally $|4m\tau(A_\tau v, v)| \le \frac{1}{2}\|A_\tau v\|^2 + C\tau^2\|v\|^2$. For large τ, the squared parts absorb the normal derivative cost and the positive zero-order term absorbs the remaining errors. This yields (5.6). Smooth first- and zero-order terms in L cost only $C(\|\nabla v\|^2 + \tau^2\|v\|^2)$ and are absorbed in the same way. The proof gives uniform constants on fixed compact sets of strict weights and bounded coefficients.

The existence criterion is a condition on a hypersurface in the product. Hessians in the following statement use the product connection of g_1 and g_2.

Proposition 5.3 (Product extension criterion)**.** *Let ρ be smooth near $z_0 = (x_0, y_0)$, with $\rho(z_0) = 0$ and $\mathrm{d}_x\rho, \mathrm{d}_y\rho \neq 0$. Put*

$$a_i = \frac{\nabla_i\rho}{|\nabla_i\rho|^2}, \qquad H = \mathrm{Hess}\,\rho.$$

Suppose that at z_0

$$H((a_1,-a_2),(a_1,-a_2))+H((v_1,v_2),(v_1,v_2))>0 \quad (v_i\perp a_i,\ |v_i|=|a_i|)\,. \tag{5.7}$$

Every product solution given on $\{\rho>0\}$ near z_0 extends to a neighborhood of z_0, agreeing on a smaller part of that side. In fact, the construction uses only two compact product sets in $\{\rho>\epsilon\}$ for some $\epsilon>0$. The neighborhoods and estimates on smaller neighborhoods are uniform under small smooth perturbations preserving the strict hypotheses, provided the data on those compact sets are bounded.

Proof. We first split the product Hessian condition into a strict weight in each factor. Their sum will lie below a defining function for the given side, with a quadratic gap. That gap places two intersecting product sets in the given side; the weights then yield interpolation exponents greater than one half, allowing Lemma 5.2 to apply.

Splitting the weight between the factors. We seek symmetric forms P_i satisfying

$$P_i(a_i,a_i)+P_i(v_i,v_i)>0, \qquad H+K\,\mathrm{d}\rho^{\otimes 2}>P_1\oplus P_2$$

for some $K>0$, with the vectors v_i as in (5.7). After division by $|a_i|^2$, the first inequality is the strict-weight condition for a function with gradient $\nabla_i\rho$ and Hessian P_i. The second inequality will give the quadratic gap. To obtain these forms by convex separation, the dual tests are positive semidefinite forms D whose diagonal blocks are

$$d_i a_i^{\otimes 2}+S_i, \qquad d_i\ge 0, \quad S_i\ge 0 \text{ on } a_i^{\perp}, \quad \operatorname{tr} S_i=d_i|a_i|^2.$$

Here is the finite-dimensional separation being used. If C is an open convex cone and $H\notin C$, fix $c_0\in C$ and put $H_e=H-ec_0$. Then $H_e\notin\overline{C}$, since $\overline{C}+C\subset C$. Let q_e be the nearest point of $\overline{C}$ to H_e, supplied by the Hilbert projection theorem. The projection inequality, tested also at 0 and $2q_e$, shows that $D_e=(q_e-H_e)/|q_e-H_e|$ satisfies $D_e:C\ge 0$ and $D_e:H_e<0$. A convergent subsequence as $e\downarrow 0$ gives $|D|=1$, $D:C\ge 0$, and $D:H\le 0$. Apply this to the cone of positive definite forms plus the allowed $P_1\oplus P_2$. The individual dual cones consist exactly of the displayed diagonal blocks: diagonalizing S_i expresses such a block as a positive sum of $a_i^{\otimes 2}+v_i^{\otimes 2}$; separation from this closed cone, followed by a small positive-definite perturbation of the separator, proves the converse. This proves the stated dual tests without an additional convex representation theorem. Normalize $\operatorname{tr} D=1$. To obtain a suitable K, it suffices by compactness to prove $H:D>0$ on tests with $D\,\mathrm{d}\rho=0$.

Represent such a D as a covariance matrix. The normal components measured by the two partial covectors are opposite. The diagonal-block condition makes each normal component uncorrelated with its own tangential component, hence with both tangential components. Their variances coincide, giving $d_1=d_2=d>0$ and

$$D=d(a_1,-a_2)^{\otimes 2}+D_\perp, \qquad \operatorname{tr}(D_\perp)_{ii}=d|a_i|^2.$$

The case $d=0$ would force $D=0$. Among nonnegative tangential forms with the two fixed traces, minimize the linear functional $H:D_\perp$ on their compact set, and choose a minimizer of least rank. If its rank is $r\ge 2$, a nonzero symmetric perturbation supported on its image preserves both traces, since $r(r+1)/2>2$. Both small signs preserve positivity, so minimality forces the objective to be constant in this direction. Continuing to an endpoint of positivity lowers the rank, a contradiction. The perturbation has trace zero and is nonzero, so has both signs and such an endpoint is finite. The minimizer therefore has rank one. Its two fixed traces give exactly the rank-one tests in (5.7), multiplied by d; their strict positivity proves $H:D>0$ for every test under consideration. Compactness of the normalized tests then supplies K, as

required. Explicitly, if no K worked, separators with trace one for $H + K\,\mathrm{d}\rho^{\otimes 2}$, as $K \to \infty$, would have a limit with $D\,\mathrm{d}\rho = 0$ and $H : D \le 0$, contradicting the positivity just proved.

Prescribe gradients $\mathrm{d}_i\rho$ and Hessians P_i for smooth functions θ_i vanishing at the respective points. In small coordinates centered there, Taylor expansion gives

$$\theta_1(x) + \theta_2(y) \le \widetilde{\rho}(x, y) - c(|x|^2 + |y|^2), \qquad \widetilde{\rho} = \rho + K\rho^2/2, \tag{5.8}$$

with $c > 0$. Each θ_i satisfies (5.5). The positive side of $\widetilde{\rho}$ is the positive side of ρ in this neighborhood.

Placing the cross and obtaining interpolation. Use coordinates (s_i, z_i) with $s_i = \theta_i$. Fix a small $\kappa > 0$ so that $\phi(s_i) = s_i - \kappa s_i^2$ remains a strict weight. Choose successively a small lateral radius R, a height $l \ll cR^2$, and $e \ll \kappa l^2$. Take

$$D_i = \{-l < s_i < l + 4e,\ |z_i| < R\}.$$

Use a cutoff equal to one on the inner lateral half and for $-l + 3e < s_i < l + 2e$, supported in the cylinder and in $-l + 2e < s_i < l + 3e$. Let $S_i \Subset D_i$ include neighborhoods of its top and lateral derivative supports, where respectively

$$s_i > l + e/2 \quad \text{or} \quad |z_i| > R/3,$$

and an inner top slice near $s_i = l + 5e/4$. The two closed product sets $\overline{D_1 \times S_2}$ and $\overline{S_1 \times D_2}$ lie strictly in the given side. For a top slice, the sum of the two s coordinates is positive. For a lateral slice, it is greater than $-2l$, which is dominated by the quadratic gap in (5.8). Making the sets slightly smaller if necessary gives a common $\epsilon > 0$ with $\rho > \epsilon$ on their closures.

For an L_i-solution normalized by $\|w\|_{L^2(D_i)} = 1$, apply (5.6) to this cutoff times w. Interior estimates bound the bottom derivative terms using the large norm and the weight $\phi(-l + 3e)$. The remaining derivative terms use $\|w\|_{L^2(S_i)}$ and a weight at most $\phi(l + 3e)$. On a smaller inner cylinder beginning at $s_i = -e$, the weight is at least $\phi(-e)$. Balancing the two exponentials yields (5.1) with

$$\gamma_i = \frac{\phi(-e) - \phi(-l + 3e)}{\phi(l + 3e) - \phi(-l + 3e)} > \frac{1}{2}. \tag{5.9}$$

Indeed at $e = 0$ the quotient is $1/2 + \kappa l/2$. Values of the balancing parameter below τ_0 are covered by increasing the constant. The output cylinder may be chosen to include a connected tube from the origin to the indicated top slice.

Constructing and identifying the extension. Lemma 5.2 now gives a product solution on a neighborhood of z_0. It agrees with the old solution there on the positive side: from a sufficiently near positive-side point, increase s_2 into the top slice. Since $\partial_{s_2}\rho > 0$ locally, the segment stays on that side and in the larger constructed cylinder. Agreement on the slice and unique continuation along a neighborhood of the segment prove agreement at the original point. All choices have strict margins. Keeping their finitely many compact sets and shrinking their output neighborhoods once proves the stated perturbation uniformity by (5.6), Lemma 5.2 and interior elliptic estimates. □

5.4 Gluing along a compact family of surfaces

Proposition 5.3 provides a germ across one surface point. The following formulation records the geometry needed to combine these germs. In particular, agreement is proved on specified overlap components, rather than inferred from the existence of a cover.

Lemma 5.4 (Propagation through a compact cylinder)**.** *Suppose a region in a product patch has smooth coordinates (z, a) near $B \times [a_-, a_+]$, where B is a compact base, possibly with boundary or corners. A product solution is given on a neighborhood of the top and lateral boundary and on an open set $\mathcal{S}$. Assume that $\mathcal{S}$ is upward closed in a at fixed z and contains these boundary neighborhoods. If every level $a =$ constant satisfies Proposition 5.3 outside $\mathcal{S}$, with the positive side pointing toward increasing a, the solution extends to a neighborhood of the cylinder, agreeing on $\mathcal{S}$. For fixed coordinates, compact sets and strict reference inequalities, the extension has uniform compact-subset bounds under small smooth perturbations, in terms of bounds on finitely many compact subsets of the initially given region.*

Proof. Starting from the top, let a_* be the infimum of levels down to which an extension agreeing with the seed has been obtained. In the uniform version include uniform bounds in this property. At a non-seed point of $B \times \{a_*\}$, Proposition 5.3 uses compact data strictly above that level. Take a finite cover of the level by such output neighborhoods and seed neighborhoods. Their data are contained above $a_* + \eta$ for one $\eta > 0$, so an already reached level supplies them with the needed bounds.

Shrink the outputs in (z, a) coordinates to boxes $B_i \times (a_* - d_i, a_* + d_i)$. Every connected component of an overlap contains vertical segments to a common height greater than a_*. Both germs equal the preceding extension there. Unique continuation for $L_1^x + L_2^y$ makes them identical on that overlap component. The same argument gives agreement with $\mathcal{S}$, because a vertical segment moving upward from a seed point stays in the seed. Boundary neighborhoods use the given solution. A finite subcover has a positive minimum output width and therefore passes the putative last level, or includes the endpoint a_-. This proves continuation on the whole cylinder. The same finite constructions retain all strict margins under small perturbations. Their Carleman, series and interior estimates prove the uniform assertion. Data bounded up to a limiting surface are never required: each local construction uses compact sets strictly above it. □

5.5 Initialization at a fixed distance from the diagonal

We apply the criterion to a mixed Green kernel. Let g_1, g_2 be smooth metrics in coordinate patches centered at 0, obtained from interior patches of compact Dirichlet manifolds. Suppose they agree on an open set W containing the lower side of a smooth hypersurface through 0. After a linear coordinate change, assume

$$g_1(0) = g_2(0) = I, \qquad \{x_n < q(x')\} \subset W \text{ near } 0, \qquad q(0) = 0, \quad \mathrm{d}q(0) = 0,$$

for a smooth function q. No regularity of the rest of ∂W is assumed. Let G_j be the Dirichlet Green kernels, normalized with metric volume, and suppose $G_1 = G_2$ on $W \times W$. For a small dilation parameter $\lambda > 0$, put

$$g_{j,\lambda}(z) = g_j(\lambda z), \qquad G_{j,\lambda}(x, y) = \lambda^{n-2} G_j(\lambda x, \lambda y), \qquad W_\lambda = \lambda^{-1} W. \tag{5.10}$$

On each fixed bounded patch the metrics tend smoothly to the Euclidean metric, and W_λ contains every fixed compact subset of $\{z_n < 0\}$ for sufficiently small λ. Initially define, off the diagonal,

$$\mathcal{G}_\lambda(x, y) = \begin{cases} G_{1,\lambda}(x, y), & y \in W_\lambda, \\ G_{2,\lambda}(x, y), & x \in W_\lambda. \end{cases} \tag{5.11}$$

The pieces fit and solve the first equation in x and the second in y.

Proposition 5.5 (Fixed-separation initialization)**.** *In this setting, let $n \geq 3$ and fix $B > 0$, $d > 0$ and $\eta > 0$. For all sufficiently small λ, the data (5.11) have a product-solution continuation to a neighborhood of*

$$\{(x, y) : |x|, |y| \leq B, \ |x - y| \geq d\}.$$

Every fixed C^m norm on a smaller neighborhood is bounded uniformly in λ. The continuation agrees with $G_{1,\lambda}$ where $y_n < -\eta$ and with $G_{2,\lambda}$ where $x_n < -\eta$, on the corresponding overlaps. The smallness threshold for λ and the constants may depend on B, d, η, m.

Proof. We first cross the limiting plane, then deform variable-radius tubes to reach the prescribed separations. All the sets and positive separation margins used in the two steps are fixed before choosing λ.

Crossing the plane and obtaining a common seed. We first extend slightly across the plane in one variable while keeping the other at a comparable positive separation. The first interpolation exponent can be made close to one; this compensates for the fixed positive exponent obtained by propagation in the separated second variable.

Fix a point x_0 of height zero and a separation scale $s > 0$. In the second factor take

$$D_2 = \{s/2 < |y - x_0| < 2s\}$$

and an observed ball S_2 about $x_0 - se_n$. In the first take a ball of radius $3\tau_0 s$ centered at $x_0 - \tau_0 s e_n$, with $\tau_0 > 0$ fixed small enough that the two large domains are separated. Its observed ball has radius $(1-\alpha)\tau_0 s$, with $\alpha > 0$ to be chosen. These observed balls lie strictly in the lower half-space.

There is a two-constants estimate from S_2 to a slightly enlarged closed annulus $4s/5 \le |y - x_0| \le 6s/5$, with an exponent $\gamma_2 > 0$ independent of α. Here is a direct way to obtain it with room for perturbations. On a Euclidean annulus the weight $|z - z_*|^{-1}$ is strict: its radial Hessian is $2|z - z_*|^{-3}$ and each tangential Hessian value is $-|z - z_*|^{-3}$. Radial cutoffs in (5.6), followed by exponential balancing, propagate smallness from an inner ball to an intermediate ball using the norm on an outer ball. A finite chain of overlapping balls with enlarged balls inside D_2 connects S_2 to the target annulus. The annulus is connected since $n \ge 3$. Multiplying the finitely many positive exponents gives the asserted γ_2. The same weights and margins work for sufficiently small perturbations of the Euclidean metric.

In the first factor use this radial argument centered at $x_0 - \tau_0 s e_n$. Keep the outer cutoff a fixed distance beyond radius $\tau_0 s$, and put the inner cutoff just inside radius $(1-\alpha)\tau_0 s$. Take a concentric target ball of radius $(\tau_0 + c)s$, with $c > 0$ small; it contains the observed ball and a ball of radius cs about x_0. As α and c decrease, the target weight approaches the inner weight. The interpolation exponent therefore tends to one. Choose these constants so that $\gamma_1 + \gamma_2 > 1$. Lemma 5.2 extends the kernel to a small ball about x_0 times the target annulus. On these fixed separated sets the rescaled Green kernels and their derivatives are uniformly bounded: the local Green singularity is $O(|x - y|^{2-n})$, and interior elliptic estimates give derivative bounds away from the pole.

The extensions agree with both lower pieces. Agreement for $y \in S_2$ comes from the first expansion. For fixed x below a sufficiently small negative height, continue in y through the connected target annulus to compare with $G_{2,\lambda}$. For a second point below that height, compare with $G_{1,\lambda}$ by continuation in the first variable from its observed ball. All comparisons remain separated.

The local extensions must next be compared on overlaps. Retain much smaller first balls and narrower annuli for coverage, keeping the larger products as comparison domains. The unused margins allow both variables to move into the known lower region. Overlapping retained products have scales comparable to $|x - y|$. For two products of the same orientation, lower their near-plane variable into the known region; the path stays inside both larger first balls. For products of opposite orientations, both variables lie in retained near-plane balls. On the fixed compact range of locations and positive separation scales, choose the retained products with strict ball and annular margins, then take $\eta_0 > 0$ much smaller than the smallest retained width. Lower both variables by

$$h = \max(x_n, y_n, 0) + 2\eta_0.$$

By making the retained balls a sufficiently small fraction of the larger balls, the path stays in both larger products: its displacement is smaller than their ball and annular margins. Its endpoint has both heights below $-\eta_0$, where both germs are the original kernel. Unique continuation along a neighborhood of the path proves agreement. Choose $\eta_0 \le \eta$ and then λ small enough for this truncated lower half-space to lie in W_λ on all sets used.

Interchanging the two variables and covering gives, for some fixed $c_0 > 0$, a single-valued continuation in

$$\min(x_n, y_n) < c_0|x - y|, \tag{5.12}$$

on every fixed compact range of bounded locations and positive separations. The estimates are uniform in λ on that range. For coverage near the plane, take $x_0 = (x', 0)$ and $s = |y - x_0|$ when the first variable has the smaller nonnegative height; then x is in the retained first ball and y in the retained annulus if c_0 is small. Points already below the plane are covered by the same construction near it or by the given cross farther below it. The preceding overlap argument makes these choices compatible.

Sweeping tubes inward from the common seed. The region (5.12) is now the seed for the second stage. At fixed center y and direction ω, we move x along the ray from y: large radii lie in the seed, and decreasing the radius will reach the target pairs. Choose $M > B + 3$ and introduce

$$t = M + y_n + e_0|y'|^2, \qquad r_a(y) = at^{1+\sigma}, \qquad x = y + r_a(y)\omega, \quad |\omega| = 1, \tag{5.13}$$

where $\sigma, e_0 > 0$ will be small. The base is

$$y_n \ge -B - 1, \qquad t \le T_0, \qquad \omega \in \mathbb{S}^{n-1},$$

and a decreases through a fixed interval $[a_{\min}, a_{\max}]$. The term $e_0|y'|^2$ makes this base compact. The parameter a decreases while (y, ω) remains fixed, as required by the compact-cylinder propagation lemma. Take $a_{\min}$ small enough that the target separations exceed $r_{a_{\min}}(y)$, and $a_{\max}$ large enough that its whole level satisfies (5.12). The bottom base boundary has $y_n < 0$. Take T_0 so large that $c_0 a_{\min} T_0^\sigma > 1$; its top boundary also satisfies (5.12). Here the superlinear power $1 + \sigma$ ensures that $r_a/t = at^\sigma$ is large on this top face. Thus the initial a-level and both base boundary faces are supplied by the seed. This known region is upward closed in a: since $\min(x_n, y_n) = y_n + r\min(\omega_n, 0)$, its defining inequality is $y_n < (c_0 - \min(\omega_n, 0))r$, whose radius coefficient is positive.

It remains to check the strict extension criterion outside that region. At the Euclidean metrics use

$$\rho = \tfrac{1}{2}(|x - y|^2 - r_a(y)^2), \qquad b = \nabla r_a.$$

Away from $\omega + b = 0$, multiply the vectors in (5.7) by the common positive factor r_a and write their real and imaginary parts as

$$\begin{aligned} U_1 &= \omega + iv, && |v| = 1, \quad v \perp \omega, \\ U_2 &= \frac{\omega + b}{|\omega + b|^2} + iw, && w \perp (\omega + b), \quad |w| = |\omega + b|^{-1}. \end{aligned} \tag{5.14}$$

The identity $\omega \cdot (U_1 - U_2) = b \cdot U_2$ cancels the normal part of the distance Hessian. Consequently the test is

$$\left|\pi_{\omega^\perp}(U_1 - U_2)\right|^2 - (r_a \partial^2 r_a)(U_2, \overline{U_2}). \tag{5.15}$$

For bounded b, outside fixed small neighborhoods of $b = -\omega$ and $b = -2\omega$, the first term is at least $c|b|^2$. To see this, a zero forces the real part of U_2 to be parallel to ω, and equality of the

projected imaginary lengths gives $|\omega + b| = 1$. Thus $b = 0$ or $b = -2\omega$. Near $b = 0$, the real projected difference controls $|\pi_{\omega^\perp} b|$ to first order; also

$$|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2),$$

so the imaginary projected difference controls $|\omega \cdot b|$ to first order. This proves the quadratic lower bound near zero; compactness proves it on the remaining range.

Outside (5.12), $y_n \geq c_0 r_a$, so $r_a/t \leq 1/c_0$ and

$$b = (1 + \sigma)(r_a/t)\nabla t$$

stays bounded. When ∇t is close to e_n, the two excluded neighborhoods already lie in (5.12). Indeed,

$$\frac{x_n}{r_a} \leq \frac{(1+\sigma)|\nabla t|}{|b|} + \omega_n,$$

which is arbitrarily small near $b = -\omega$ and negative near $b = -2\omega$ as σ and the gradient error tend to zero. On the complement $|U_2|$ is bounded and differentiation of (5.13) gives

$$r_a |\partial^2 r_a| \leq C(\sigma + T_0 e_0)|b|^2. \tag{5.16}$$

Choose the excluded-neighborhood sizes first in terms of c_0. Use the positive lower bound on the remaining bounded b-range to choose σ small, then choose the a-interval and T_0 as above, and finally $e_0 > 0$ so small that (5.16) is absorbed. Choosing the transverse compactification last keeps its contribution to the Hessian test small. These choices also make the gradient error small, since $|\nabla t - e_n| \leq 2\sqrt{e_0 T_0}$ on the base. Thus (5.15) is strictly positive wherever continuation is needed. Both partial gradients are nonzero there.

All these sets are bounded and stay at separation at least $a_{\min}(M - B - 1)^{1+\sigma} > 0$. Smooth convergence of the metrics therefore preserves the strict tests for sufficiently small λ. Lemma 5.4 continues the kernel through the cylinder, with uniform estimates and agreement on the seed. A target pair has $a = |x - y|/t^{1+\sigma} \geq a_{\min}$; if $a > a_{\max}$ it is already in the coarse region. Hence every target pair is covered.

We also identify the extension with both pieces of the full cross (5.11). Perform the construction with a larger bound $B' > B$ and a smaller separation $0 < d' < d$, chosen so that for every $|y| \leq B$ the domain

$$D_y = B(0, B') \setminus \overline{B(y, d')}$$

is connected and contains a fixed lower open ball with $x_n < -\eta$. Choose B' large enough that every excluded closed ball lies strictly inside $B(0, B')$ and is disjoint from that lower ball. For fixed $y \in W_\lambda$ in the target range, the continuation and $G_{1,\lambda}$ solve the same first-variable equation on D_y. On the lower ball the continuation equals $G_{2,\lambda}$, which equals $G_{1,\lambda}$ because both variables lie in W_λ. Unique continuation on D_y proves agreement on the required overlap. Interchanging the variables proves agreement with the other piece. Keeping slightly larger target bounds and a slightly smaller separation before restricting to the prescribed range gives the claimed neighborhood and its compact-subset estimates. □

The constants in Proposition 5.5 may deteriorate as $d \downarrow 0$. Its role is to initialize a construction at finitely many fixed positive scales. Lemma 5.4 will also provide qualitative existence at each positive radius in a shrinking family; the uniform estimate for that family is established below.

6 Continuation to shrinking spheres

We now apply the product continuation criterion near a point where an isometry is already known on one side. The purpose of this section is to construct a family of functionals on harmonic functions, represented on small spheres. Their existence will follow from the geometry of the spheres. A uniform bound for the functionals, needed when the spheres shrink, will follow from the estimate in the next section.

6.1 The local data and their normalization

Throughout Sections 6–8, fix the manifolds (N_j, g_j), actual Dirichlet Green functions G_j, common harmonic coordinate neighborhood Ω, matching set W, and finite harmonic pairs (u_1^a, u_2^a) of Theorem 5.1. Thus the coordinate metrics and paired functions agree on W, and the Green kernels agree on $W \times W$ off the diagonal. The first-side coordinate Hessians span the hyperplane annihilated by $g_1^{-1}(0)$. An interior ball in W tangent at 0 supplies the one-sided hypersurface required by that theorem. We construct the sphere functionals for these data.

A linear change of harmonic coordinates makes $g_1(0) = g_2(0) = I$ and writes the separating hypersurface as a graph tangent to $x_n = 0$, with the known side below it. For a spatial dilation $\lambda > 0$, write

$$g_{j,\lambda}(x) = g_j(\lambda x), \qquad G_{j,\lambda}(x, y) = \lambda^{n-2} G_j(\lambda x, \lambda y). \tag{6.1}$$

Dilate the harmonic pairs at the same time:

$$u_{j,\lambda}^a(x) = u_j^a(\lambda x).$$

They remain harmonic for $g_{j,\lambda}$ and agree on the dilated known set. The metrics and Green kernels in (6.1) have the corresponding normalization on the dilated neighborhoods. On each fixed bounded coordinate set the metrics converge smoothly to I as $\lambda \downarrow 0$. The dilated known set contains every fixed compact set lying strictly below $x_n = 0$, once λ is sufficiently small.

Use x for the first variable and y for the second. The mixed kernel $\mathcal{G}$ is initially $G_{1,\lambda}$ when y is in the known set, and $G_{2,\lambda}$ when x is in that set. These prescriptions agree on their overlap and solve the first equation in x and the second in y, away from the diagonal. Proposition 5.5 continues this kernel to any specified compact region of fixed positive separation, with smooth bounds uniform for small λ. We shall use that result only after all the positive separation ranges in question have been fixed.

The geometry of a moving sphere dictates the next choices. In the strict extension test, the leading terms involving the gradient of its radius cancel; strict concavity of the logarithmic radius supplies the remaining positive term (see (6.12)). We therefore choose a conformal reference metric for which a depth function is strictly concave. Separately, a scalar normalization of the second inverse metric will give the two inverse matrices the same trace relative to the reference metric. The harmonic Hessians will determine their remaining difference.

Fix the coordinate cylinder

$$Y = \{y = (y', y_n) : |y'| \le 1,\ -1 \le y_n \le 1\}, \qquad t_0(y) = y_n + |y'|^2.$$

Take a sufficiently large fixed number L and put

$$\gamma = e^{-2Lt_0} g_{1,\lambda}, \qquad Q = \frac{n\, g_{2,\lambda}^{-1}}{\operatorname{tr}(\gamma g_{2,\lambda}^{-1})}, \qquad h = Q - \gamma^{-1}. \tag{6.2}$$

Thus Q is the principal matrix of a positive multiple of the second equation and $\operatorname{tr}(\gamma h) = 0$. The first equation can likewise be multiplied by a positive function to give principal metric γ. Such multiplication does not change its solutions. The metric γ will also identify the two tangent spaces at a common coordinate point. All these coefficients have uniform smooth bounds on a fixed neighborhood of Y for sufficiently small λ, and $h \to 0$ in every fixed smooth norm as $\lambda \downarrow 0$.

Define the depth function

$$t(y) = \int_{1/8}^{t_0(y)} e^{-2Ls}\, ds. \tag{6.3}$$

It is negative at 0 and has nonvanishing differential on Y. The conformal connection formula gives

$$\operatorname{Hess}_\gamma t = e^{-2Lt_0}\big(\operatorname{Hess}_{g_{1,\lambda}} t_0 - L|dt_0|^2_{g_{1,\lambda}} g_{1,\lambda}\big) \le -c\gamma. \tag{6.4}$$

Indeed $|dt_0|$ is bounded below, whereas the first Hessian on the right is uniformly bounded. Choose L first and then an upper bound for λ so that the displayed strict inequality holds with fixed $c > 0$.

6.2 Radii and cutoff regions

For a fixed integer $p \geq 2$, a positive amplitude R_*, and $0 < \delta \leq 1$, define

$$\ell_\delta(t) = \delta \log(1 + e^{t/\delta}), \qquad r_\delta(y) = R_* \ell_\delta(t(y))^p, \qquad f_\delta = \log r_\delta, \quad \beta_\delta = |df_\delta|_\gamma. \tag{6.5}$$

The balls will be the γ-geodesic balls with center y and radius $r_\delta(y)$, lying in the first coordinate neighborhood. As the lemma below shows, they collapse where $t \leq 0$, including a neighborhood of the contact point, but their radii stay bounded below where t is bounded away from zero on the positive side. This distinction will keep cutoff derivatives away from uncontrolled shrinking spheres.

Lemma 6.1 (Profile bounds)**.** *On Y, after the preceding choices of geometry, the following bounds hold uniformly for small λ, $0 < \delta \leq 1$, and $p \geq 2$:*

$$\beta_\delta \geq cp, \qquad \operatorname{Hess}_\gamma f_\delta \leq -c\beta_\delta \gamma, \qquad |\partial^j f_\delta| \leq C_j \beta_\delta^j \quad (j \geq 1). \tag{6.6}$$

The constants c, C_j do not depend on p. The functions $r_\delta^{1/p}$ have a Lipschitz bound independent of δ and λ for fixed p, R_*. The radii increase with δ, tend to zero on $\{t \leq 0\}$ as $\delta \downarrow 0$, and satisfy

$$\sup_{\delta, y} r_\delta \beta_\delta^2 \longrightarrow 0 \quad \text{as } R_* \downarrow 0 \tag{6.7}$$

for each fixed p.

Proof. Write $a_\delta = (\log \ell_\delta)'$. In terms of $s = t/\delta$,

$$a_\delta(t) = \delta^{-1} A(s), \qquad A(s) = \frac{e^s}{(1 + e^s)\log(1 + e^s)}.$$

The function A is positive and decreasing. At the negative end it tends to one, and at the positive end it is comparable to $(1+s)^{-1}$. Differentiating the formula, or its convergent expressions at the two ends, gives $|A^{(j)}(s)| \leq C_j A(s)^{j+1}$. On the remaining compact interval the same inequalities follow by positivity. On the fixed t interval this gives a positive lower bound for a_δ and

$$a_\delta' \leq 0, \qquad |\partial_t^j \log \ell_\delta| \leq C_j a_\delta^j, \qquad a_\delta \leq \ell_\delta^{-1}.$$

Since $|dt|_\gamma$ is bounded above and below, $\beta_\delta = p a_\delta |dt|_\gamma$. Equation (6.4) and

$$\operatorname{Hess}_\gamma f_\delta = p a_\delta \operatorname{Hess}_\gamma t + p a_\delta' \, dt \otimes dt$$

prove the Hessian bound. The chain rule proves the other derivative bounds, with constants independent of $p \geq 2$ because $p a_\delta^j \leq p^j a_\delta^j$.

The bound $0 < \ell_\delta' < 1$ gives the asserted Lipschitz estimate for $r_\delta^{1/p} = R_*^{1/p} \ell_\delta \circ t$. Moreover,

$$\partial_\delta \ell_\delta = \log(1 + e^s) - \frac{s e^s}{1 + e^s} > 0.$$

The function on the right increases up to $s = 0$ and decreases afterward, with limit zero at both ends. The pointwise limit of ℓ_δ is $\max(t, 0)$. Finally,

$$r_\delta \beta_\delta^2 \leq C R_* p^2 \ell_\delta^{p-2},$$

and ℓ_δ is uniformly bounded on the fixed interval. This proves (6.7). □

Choose $t_b > 0$ so small that $3t_b < t(1/4)$, where on the right t denotes the increasing function of t_0 in (6.3), and set

$$K = \{y \in Y : y_n \geq -1/2,\ t(y) \leq 2t_b\}. \tag{6.8}$$

Then $K \Subset Y^\circ$. We fix a smooth cutoff $\chi \in C_c^\infty(Y^\circ)$ equal to one on a neighborhood of K, with

$$\operatorname{supp}(d\chi) \subset \{y_n < -1/4\} \cup \{t > t_b\}. \tag{6.9}$$

For example, multiply a cutoff that changes from zero to one between $y_n = -3/4$ and $y_n = -1/2$ by a cutoff of t that changes from one to zero between $2t_b$ and $3t_b$, moving the endpoints slightly to make it one on a neighborhood of K. The inequality $3t_b < t(1/4)$ keeps its support away from the lateral and upper faces of Y. This gives fixed room around K and places every cutoff derivative either in the known lower region or where

$$r_\delta \geq R_* t_b^p > 0.$$

Thus the later estimates for the sphere functionals can use the original Green kernel or fixed-separation continuation on every cutoff derivative support.

6.3 The strict test at a moving sphere

For coordinate derivatives of the tensor h, use the pointwise notation

$$H_8(y) = \sum_{|\alpha| \leq 8} |\partial_y^\alpha h(y)|.$$

We shall temporarily impose

$$H_8 \leq \varepsilon r_\delta, \qquad r_{\delta'} \beta_{\delta'}^2 \leq \varepsilon \quad (\delta \leq \delta' \leq 1) \quad \text{on } Y. \tag{6.10}$$

The first inequality measures how closely the two principal metrics agree compared with the current radius. It will be improved in Section 8.

Proposition 6.2 (Continuation to the sphere family)**.** *Choose p sufficiently large and then $\varepsilon > 0$ sufficiently small in terms of the fixed background bounds. Choose $R_* > 0$ small enough that all the balls lie in normal neighborhoods and (6.7) is at most ε. For all sufficiently small spatial dilations λ, and every $0 < \delta \leq 1$ for which (6.10) holds, the mixed kernel extends to a neighborhood of*

$$\{(x, y) : y \in Y,\ d_\gamma(x, y) = r_\delta(y)\}.$$

The extension solves the two homogeneous equations and agrees with the prescribed lower cross and the fixed-separation continuation. It is smooth for each positive δ; no uniform norm as $\delta \downarrow 0$ is asserted here.

Proof. We check Proposition 5.3 at each intermediate radius $r = r_{\delta'}$. Put $f = \log r$, $\beta = |df|_\gamma$, and $b = \nabla^\gamma r$, so $|b| = r\beta$. Use the defining function

$$\rho(x, y) = \tfrac{1}{2}\big(d_\gamma(x, y)^2 - r(y)^2\big).$$

The positive side is the outside of the sphere. Parallel translation along its radial geodesic identifies the two tangent spaces for γ; write ω for the unit vector from y to x.

In the normalization of the strict test, the normal and tangential vectors can be combined into complex vectors

$$U_1 = \omega + iv, \quad v \perp \omega, \quad |v| = 1, \qquad U_2 = q + iw,$$

where the conditions for the second principal metric Q^{-1} give

$$(\omega + b) \cdot U_2 = 1, \qquad q = \frac{\omega + b}{|\omega + b|^2} + O(|h|), \qquad |w| = |\omega + b|^{-1} + O(|h|). \tag{6.11}$$

Here and below lengths and dot products in these formulas use γ and its parallel identification. The first equation includes $(\omega + b) \cdot w = 0$. Since $|b|$ and $|h|$ are small, both partial gradients of ρ are nonzero and $|U_2|$ is bounded above and below.

The Hessian of half the squared distance, with the product γ-connection, evaluated on these endpoint vectors is

$$|U_1 - U_2|^2 + O(r^2)(|U_1|^2 + |U_2|^2).$$

To see the error size, parametrize the radial geodesic on $[0, 1]$. Its Jacobi field with prescribed endpoint values differs, in a parallel frame, from the affine interpolant by $O(r^2)$ in C^1, by the Jacobi equation and bounded curvature. The second variation formula then gives the displayed bound. Replacing the second connection by that of Q^{-1} costs $O(r^2)$ as well: the connection difference is $O(|h| + |\partial h|) = O(\varepsilon r)$ and $|d\rho| = O(r)$.

Now $\omega \cdot (U_1 - U_2) = b \cdot U_2$ and $\operatorname{Hess}(r^2/2) = 2\, dr \otimes dr + r^2 \operatorname{Hess} f$. Consequently the strict Hessian expression is

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - |b \cdot U_2|^2 - r^2 \operatorname{Hess}_\gamma f(U_2, \overline{U}_2) + O(r^2). \tag{6.12}$$

For a real Hessian, its value on a complex vector and its conjugate is the sum of its values on the real and imaginary parts, as required by the criterion.

The real part of the projected difference is $-\pi_{\omega^\perp} b + O(|b|^2 + |h|)$. For its imaginary part, (6.11) gives $|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2 + |h|)$; compare this with $|v| = 1$. Squaring these two bounds yields

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 \geq |b|^2 - C(|b|^3 + |b||h| + |h|^2).$$

The real and imaginary directions of U_2 are almost orthonormal, so

$$|b \cdot U_2|^2 \leq (1 + C(|b| + |h|))|b|^2.$$

By Lemma 6.1, the expression (6.12) is therefore bounded below by

$$cr^2\beta - C\big(r^2 + r^3\beta^3 + \varepsilon r^2\beta + \varepsilon^2 r^2\big).$$

Choose p so that $\inf \beta$ absorbs the Cr^2 term. Then choose ε small. The inequality $r\beta^2 \leq \varepsilon$ absorbs $r^3\beta^3$, and the expression is strictly positive.

It remains to specify the starting and boundary data for this local test. Deform δ' from 1 to δ, using (y, ω) as base variables. The inequality $H_8 \leq \varepsilon r_\delta$ implies its counterpart for every $\delta' \geq \delta$, because the radii increase with δ'. At $\delta' = 1$ the required separations have a fixed positive minimum. The same is true wherever $t > t_b$, since $r_{\delta'} \geq R_* t_b^p$. Proposition 5.5 supplies uniformly bounded data on these fixed separation ranges, with open room. Where $y_n < -1/4$, use the original $G_{1,\lambda}$; for small λ this is part of the known lower cross. These regions cover the lateral and end boundaries of the cylinder that are not supplied by the initial parameter, and their prescriptions agree on overlaps.

The base sphere bundle over the closed cylinder is compact. On each fixed interval $\delta \leq \delta' \leq 1$, the radii are positive, all surfaces lie in normal neighborhoods, and the strict tests just proved apply away from the prescribed regions. Lemma 5.4 therefore supplies compatible extension germs on the whole family. Its finite-cover gluing also retains agreement with the prescribed regions. This is an existence argument for each finite interval; it does not assert uniform continuation bounds when the lower endpoint tends to zero. □

6.4 Functionals represented on the spheres

The mixed kernel now defines a map from first-metric harmonic functions to second-metric harmonic functions. Its representation on a small sphere will allow us to estimate this map without estimating the kernel through every continuation step. Fix $0 < \delta \le 1$ and write $r = r_\delta$ in this subsection.

For a function u harmonic for $g_{1,\lambda}$ near $\overline{B_\gamma(y, r(y))}$, define

$$T_y u = \int_{\partial B_\gamma(y,r(y))} \left(\mathcal{G}(x,y)\partial_{\nu_1} u(x) - u(x)\partial_{\nu_1}\mathcal{G}(x,y)\right) dS_1(x). \tag{6.13}$$

Here ν_1 is the outer unit normal and dS_1 the hypersurface measure for the actual first metric $g_{1,\lambda}$, not for γ. Let S_y be the unit sphere of $(T_y\mathbb{R}^n, \gamma_y)$, with its rotation-invariant probability measure $d\sigma_y$. Parametrize the integration sphere by $x = \exp_y^\gamma(r(y)\omega)$.

Proposition 6.3 (The transfer density)**.** *Under the assumptions of Proposition 6.2, there is a smooth function $\psi(y,\omega)$ on the unit sphere bundle over Y° such that*

$$T_y u = \int_{S_y} \psi(y,\omega) u\big(\exp_y^\gamma(r(y)\omega)\big)\, d\sigma_y(\omega). \tag{6.14}$$

For every fixed center y_0, if u is first-metric harmonic on a neighborhood of $\overline{B_\gamma(y_0, r(y_0))}$, then $y \mapsto T_y u$ is defined and second-metric harmonic on a sufficiently small neighborhood of y_0. This neighborhood may depend on u. In particular, for functions harmonic on the fixed coordinate neighborhood required by the sphere family,

$$\int_{S_y} \psi\, d\sigma_y = 1, \qquad T_y x^i = y^i, \qquad T_y u^a_{1,\lambda} = u^a_{2,\lambda}(y). \tag{6.15}$$

On the support of derivatives of χ, every fixed angular smooth norm of ψ and its first base derivatives is bounded independently of δ and small λ, after p, R_* and the fixed geometry have been chosen.

Proof. For each ball, its smooth Dirichlet solution operator determines the normal derivative of a harmonic function from its boundary trace. Move this boundary operator in the first term of (6.13) onto the smooth boundary value of $\mathcal{G}$ by adjunction, and then pull back to S_y. The resulting smooth density is ψ. Smooth dependence of the Dirichlet problem and of the sphere parametrization gives its smooth dependence on y for each positive radius. This construction initially applies to smooth boundary values and hence to every harmonic function used in (6.14).

Fix a center y_0 and a function u harmonic on a neighborhood of its closed ball. The continued kernel is defined on an open neighborhood of the compact sphere over y_0. Choose a fixed slightly larger surrounding surface inside this kernel neighborhood and the domain of u. After restricting the center neighborhood, it surrounds every moving sphere and the whole intervening collar remains in that common domain. Green's identity on the collar replaces the moving integral in (6.13) by the integral over this fixed surface, since both first-variable equations are homogeneous there. Applying the second-variable operator under the fixed integral proves the stated local harmonicity. This argument uses arbitrary local harmonic functions, not only the source-generated pairs.

On the lower known region $\mathcal{G} = G_{1,\lambda}$, so Green's formula gives $T_y u = u(y)$. Unique continuation on the connected set Y° now proves all identities in (6.15), first for the constant and coordinate pairs and then for each given harmonic pair.

For the uniform bounds on the cutoff region, distinguish the two sets in (6.9). On the known lower set, T_y is ordinary evaluation. Its density is the Poisson density at the center of the ball, expressed in normalized coordinates. After pulling the equation back by $z \mapsto \exp_y^\gamma(rz)$ and multiplying by the Laplacian scaling, the coefficients form a smooth uniformly elliptic

family down to $r = 0$, whose limiting equation is the Euclidean Laplace equation. The interior evaluation kernel for the Dirichlet problem consequently has uniform angular smooth bounds and smooth center–radius dependence. This follows directly by differentiating the Dirichlet equation and using its uniform elliptic estimates; the center stays a fixed positive distance from the unit sphere. The chain rule for a base derivative costs $|dr| = r\beta$, which is uniformly bounded by (6.10).

On the other cutoff set, $t > t_b$. There the radii have a fixed positive lower bound and all their required derivatives are bounded uniformly in δ. The fixed-separation bounds of Proposition 5.5, followed by the smooth Dirichlet construction of ψ, give the same conclusion. Compactness of the cutoff derivative supports completes the proof. □

Figure 3 records the transfer and the two parameter roles. The next section establishes a bound for its density that remains uniform as the spheres shrink.

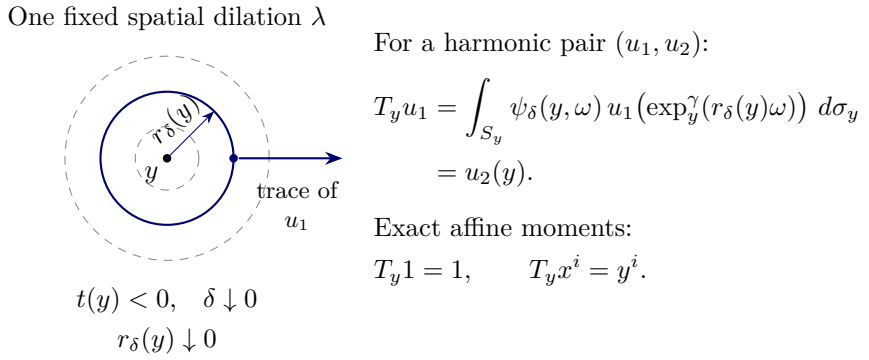


Figure 3: A normal-coordinate section of γ-geodesic spheres in the first variable, where γ is the conformal reference metric. The center y satisfies $t(y) < 0$. The spatial dilation λ is fixed while δ decreases. The Green functional transfers paired harmonic functions and reproduces affine harmonic coordinates exactly. The unit-sphere measure $d\sigma_y$ has total mass one; the density ψ_δ need not be positive.

7 An estimate on the shrinking spheres

The functional in Proposition 6.3 is defined at every positive radius. We now estimate its representing density uniformly as the radius shrinks. The equation for this density has second-order angular and base terms. Comparing a degree-raising operator with a degree-lowering operator will control both kinds of derivatives; a second comparison will retain the principal terms caused by the changing metric.

7.1 The equation for the density

We use the reference metric γ, the normalized second principal matrix Q, and $h = Q - \gamma^{-1}$ from (6.2). In this section the subscript δ on the positive radius r is suppressed. Put

$$f = \log r, \qquad \beta = |df|_\gamma.$$

The required profile bounds are

$$\beta \geq \beta_0, \qquad \mathrm{Hess}_\gamma f \leq -c_0 \beta \gamma, \qquad |\nabla^j f| \leq C_j \beta^j \quad (1 \leq j \leq 8), \tag{7.1}$$

where $c_0 > 0$ and C_j are fixed. Lemma 6.1 provides these bounds with β_0 arbitrarily large by choosing p sufficiently large. All background metric bounds below are on a fixed relatively compact coordinate region; their constants are uniform in the spatial dilation.

Let $S_\gamma Y$ be the bundle of γ-unit tangent spheres over the base region Y. Each sphere has probability measure $d\sigma_y$. Horizontal differentiation ∇ uses parallel transport for γ, including the covariant tensor indices of an operator or section. This connection preserves the sphere measure. Unless otherwise specified, all norms and full adjoints use $dV_\gamma(y)\, d\sigma_y(\omega)$.

On each tangent unit ball, extend a sphere function harmonically for the Euclidean metric $\gamma(y)$. Denote the boundary radial derivative of this extension by B, and its ambient derivatives in an orthonormal frame by A_i. Thus $B = k$ on the space $\mathcal{H}_k$ of spherical harmonics of degree k, and $A_i : \mathcal{H}_k \to \mathcal{H}_{k-1}$. The star in A_i^* denotes the sphere adjoint. These operators are parallel, with the indicated tensor indices, and

$$[B, A_i^*] = A_i^*, \qquad \sum_i A_i^* A_i^* = 0. \tag{7.2}$$

Indeed differentiation lowers homogeneous degree by one, and the second identity is the adjoint of the vanishing Laplacian of a harmonic extension. For $s \in \mathbb{R}$, use the fiber Sobolev norm

$$\|v(y,\cdot)\|_s = \|(1+B)^s v(y,\cdot)\|_{L^2(S_y)}.$$

Lemma 3.1 gives the angular pseudodifferential mapping and form bounds in this norm, including the half orders used below. Tensor-valued versions use the sum over an orthonormal frame.

We write Ψ^m for the classical pseudodifferential operators of order m on the sphere. For a positive function $a(y)$, the notation $a\Psi^m$ means that, after division by a, the finitely many symbol and operator seminorms used below are bounded. A statement with j base derivatives permits the indicated additional factor β^j.

Lemma 7.1 (Moving trace operators)**.** *Suppose that the normal balls and the functional T_y of Proposition 6.3 are defined, and that*

$$\sum_{|\alpha|\le 8} |\partial_y^\alpha h| \le \varepsilon r, \qquad r\beta^2 \le \varepsilon. \tag{7.3}$$

If u is first-metric harmonic near one such ball, set

$$U(y,\omega) = u\big(\exp_y^\gamma(r(y)\omega)\big).$$

There are sphere operators $\mathcal{C}_i$ such that $\nabla_i U = \mathcal{C}_i U$. Their sphere adjoints satisfy

$$\mathcal{C}_i^* = r^{-1} A_i^* + f_i B + E_i + F_i, \qquad E_i \in r\Psi^1, \quad F_i \in \Psi^0, \qquad f_i = \nabla_i f. \tag{7.4}$$

The normalized remainders $r^{-1}E_i$ and F_i have uniform symbol bounds after up to six base derivatives, with the allowance $C_j\beta^j$.

Write the normalized second equation, in the reference connection, as $Q^{ij}\nabla_i\nabla_j + b_2^i\nabla_i$. Its drift b_2 is uniformly controlled. If $\psi(y,\omega)$ represents T_y against $d\sigma_y$, then

$$P\psi = 0, \qquad P = r\left[Q^{ij}(\nabla_i + \mathcal{C}_i^*)(\nabla_j + \mathcal{C}_j^*) + b_2^i(\nabla_i + \mathcal{C}_i^*)\right]. \tag{7.5}$$

The products in this formula are covariant products. In particular, Q is not differentiated by taking a base adjoint.

Proof. Normalize the principal part of the first equation to γ^{-1} and pull it back by $z \mapsto \exp_y^\gamma(rz)$ to the unit ball. After multiplication by r^2, its principal coefficients are $\delta^{ij} + O(r^2)$ and its first-order coefficients are $O(r)$. The resulting smooth family remains uniformly elliptic and has an invertible Dirichlet problem, including at $r = 0$. Its radial derivative operator therefore has the expansion

$$B + rB_1(y) + r^2 B_2(y,r), \qquad B_1 \in \Psi^0, \quad B_2 \in \Psi^1. \tag{7.6}$$

Here is the parameter justification. The classical boundary-symbol construction is described in [2, Theorem 1.1(i) and Proposition 3.1]; we keep the center and radius dependence explicitly. In boundary coordinates the decaying normal root of the principal symbol constructs the Dirichlet parametrix. Successive lower-order transport equations construct its symbols smoothly in (y, r). The first normal derivative trace is an operator of order one, whose principal symbol depends only on the principal coefficients and the differentiating vector. Those coefficients have zero first r-derivative at $r = 0$, so that derivative of the trace operator has order zero. Taylor's formula in this symbol class gives (7.6) to smoothing error. The true solution differs from the parametrix by the Dirichlet inverse of a smooth error; differentiated Dirichlet estimates control this remainder in every fixed seminorm. More explicitly, on a compact center set the rescaled Dirichlet operator is smoothly invertible between each fixed pair of Sobolev spaces, uniformly for $0 \le r \le r_0$. Differentiating that inverse controls every fixed number of parameter derivatives of the smoothing correction; increasing the Sobolev orders controls its smooth kernel seminorms. Thus the Taylor expansion holds in the complete classical symbol topology, including the smoothing remainder, and not merely in one Sobolev operator norm. This proves the stated expansion and its parameter bounds.

Differentiate the exponential map in its center, transporting ω parallel to the center velocity. The Jacobi field has initial derivative zero. After inverting the differential in the z variable, the velocity acting on the unit-ball solution is

$$r^{-1}(e_i + O(r^2)) + f_i\omega.$$

The coefficient error is smooth in (y, r, ω). Tangential derivatives act directly on the boundary value; the radial derivative is (7.6). At $r = 0$ the first term is $r^{-1}A_i$. The order-one errors are $O(r)$, and the order-zero errors are bounded: in particular rf_iB_1 is bounded because $r\beta \le \varepsilon/\beta$. Taking sphere adjoints gives (7.4). Differentiating the radius costs at most one factor β per derivative, by (7.1); the asserted parameter estimates follow.

For fixed u, differentiate $T_y u = \langle \psi(y, \cdot), U(y, \cdot)\rangle_{S_y}$. Metric compatibility and $\nabla_i U = \mathcal{C}_i U$ put the differentiated operator on ψ as $\nabla_i + \mathcal{C}_i^*$. The local second-metric harmonicity in Proposition 6.3 proves that (7.5) pairs to zero with the trace of every function harmonic near the fixed closed ball. No global extension or source-family approximation is used in this step. At a fixed sphere these traces are dense among smooth boundary data: solve the smooth Dirichlet problem on slightly enlarged balls, using boundary values transported from the limiting sphere, and use smooth dependence of the solution on the radius. Thus the paired equation is the stated equation for ψ. □

7.2 The model operator and its principal perturbations

We separate a degree-raising model operator from the remaining principal and lower-order terms of the moving trace equation. We continue under the hypotheses of Lemma 7.1, including (7.3).

Define

$$D_i^+ = \nabla_i + f_iB, \qquad D_i^- = -\nabla_i + f_iB, \qquad T = \sum_i A_i^* D_i^+, \quad T^\flat = \sum_i A_i D_i^-. \tag{7.7}$$

The symbol T here denotes a differential operator on sphere functions; the transfer functional continues to be denoted T_y. Expand (7.5) by (7.4). The identities (7.2) give

$$D_i^+(r^{-1}A_j^*) = r^{-1}A_j^* D_i^+, \qquad \sum_i A_i^* A_i^* = 0.$$

Thus, in ordinary base coordinates and an orthonormal sphere trivialization,

$$P = 2T + U + J, \tag{7.8}$$

where J is a uniformly bounded family in Ψ^1, with no base derivatives, and

$$U = \sum_{|\alpha|\le 2} U_\alpha(y)\partial_y^\alpha, \qquad U_\alpha \in \varepsilon\beta^{-|\alpha|}\Psi^{2-|\alpha|}. \tag{7.9}$$

Up to five derivatives of these coefficients have the additional allowance $C_j\beta^j$. The second-base-derivative term is precisely

$$rQ^{ij}\partial_{y_i}\partial_{y_j}; \tag{7.10}$$

its coefficients are real scalars independent of the sphere variable.

We detail these bounds. The pure angular second derivative contracted with γ^{-1} vanishes. The remaining contraction is $(h^{ij}/r)A_i^*A_j^*$, and belongs to the zero-base-derivative term of (7.9). The mixed correction is $2h^{ij}A_i^*D_j^+$ and has the stated small bounds. The D^+D^+ term has coefficient r; its base orders two, one and zero have coefficients $O(r)$, $O(r\beta)$ and $O(r\beta^2)$, respectively. The condition $r\beta^2 \le \varepsilon$ gives (7.9) for all three. Terms containing E_i obey the same bounds. Terms containing F_i also do so, except for their cross terms with $r^{-1}A_j^*$, which are bounded angular order-one terms and belong to J. The drift term with A_i^* belongs to J as well. Differentiating any of these expressions gives the stated coefficient bounds by (7.1) and (7.3). A horizontal connection written in this trivialization adds a bounded angular first derivative, which preserves the same estimates. The bounds for J use the fixed background and remain uniform as β_0 is increased.

The decomposition identifies the model operator $2T$ and the second-order perturbation U. To carry the model estimate over to P we shall need more than an ordinary first-derivative estimate: the error forms also contain mixed base and angular derivatives. For a sphere function φ, put

$$\mathcal{N}(\varphi)^2 = \int_Y \left(\|\nabla\varphi\|_0^2 + \beta^2\|\varphi\|_1^2 + \beta^{-1}\|\nabla\varphi\|_{1/2}^2 + \beta\|\varphi\|_{3/2}^2\right) dV_\gamma. \tag{7.11}$$

The last two terms of $\mathcal{N}$ place half an angular derivative on $\nabla\varphi$ and three halves on φ. They will bound the remaining error forms left by the comparison with P.

7.3 A comparison of raising and lowering operators

The next estimate supplies these two derivative strengths for the model operator. Spherical harmonics are trace-free symmetric tensors, and the calculation below compares their raising and lowering operators. Here the negative Hessian of the spatial weight supplies the positive term that absorbs the bounded curvature. We give the calculation with its degree factors.

Lemma 7.2 (Weighted raising estimate)**.** *Let $n \ge 3$, and suppose (7.1) holds on a fixed base region with uniformly bounded reference geometry. There are β_0 and $c > 0$, depending only on that geometry and the constants in (7.1), such that every smooth, compactly base-supported function with zero fiber mean satisfies*

$$\|T\varphi\|^2 - \|T^\flat\varphi\|^2 \ge c\mathcal{N}(\varphi)^2. \tag{7.12}$$

Proof. Both operators change angular degree by exactly one, so it suffices to prove the estimate at one input degree $k \ge 1$. Identify that component with a harmonic homogeneous polynomial $v(z)$ of degree k. Use the Fischer inner product, in which the monomials z^α are orthogonal with squared norm $\alpha!$. Polynomial differentiation $a_i = \partial_{z_i}$ is adjoint to multiplication $M_i = z_i$. On harmonic polynomials of degree k the Fischer squared norm is the normalized squared sphere norm multiplied by

$$N_k = n(n+2)\cdots(n+2k-2), \qquad N_0 = 1.$$

For completeness, integration against a standard real Gaussian identifies the Fischer norm on trace-free homogeneous polynomials with their Gaussian L^2 norm; The Gaussian generating

function is $\mathbb{E}e^{t\cdot X} = e^{|t|^2/2}$. Its coefficient formula pairs polynomial indices. A pairing within either trace-free polynomial is zero, so only cross-pairings remain, yielding $\mathbb{E}[v(X)\overline{w(X)}] = \sum_\alpha \alpha! v_\alpha \overline{w_\alpha}$. The radial moment $\mathbb{E}|X|^{2k} = n(n+2)\cdots(n+2k-2)$ then gives the factor N_k.

Set $d = k + n/2 - 1$ and use $D_i^\pm = \pm\nabla_i + k f_i$ at this fixed degree. For harmonic polynomials q_i of degree k, the Laplacian of $\sum_i M_i q_i$ is $2\sum_i a_i q_i$. Since

$$\Delta_z(|z|^2 s) = 4d\, s \qquad (s \in \mathcal{H}_{k-1}),$$

its harmonic projection is $\sum_i M_i q_i - |z|^2 \sum_i a_i q_i/(2d)$. Orthogonality of this projection and its complement shows that its squared norm is

$$\sum_i \|q_i\|^2 + \sum_{i,j} (a_j q_i, a_i q_j) - d^{-1}\Big\|\sum_i a_i q_i\Big\|^2.$$

Changing from Fischer adjoints to sphere adjoints gives

$$A_i^*|_{\mathcal{H}_k} = \frac{N_{k+1}}{N_k}\Pi_{k+1} M_i,$$

where Π_{k+1} is harmonic projection in homogeneous degree $k+1$. Consequently N_k times the left side of (7.12) is

$$(2d+2)\left[\|D^+ v\|^2 + \sum_{i,j}(a_j D_i^+ v, a_i D_j^+ v) - \frac{1}{d}\Big\|\sum_i a_i D_i^+ v\Big\|^2\right] - 2d\Big\|\sum_i a_i D_i^- v\Big\|^2. \tag{7.13}$$

All norms in this calculation include integration over the base and use the polynomial inner product in the fibers. The factors are $N_{k+1}/N_k = 2d+2$ and $N_k/N_{k-1} = 2d$.

Write $C_{ij} = 2k(\mathrm{Hess}_\gamma f)_{ij}$ and

$$C[av] = \int_Y \sum_{i,j} C_{ij}\langle a_i v, a_j v\rangle\, dV_\gamma.$$

Covariant integrations by parts give

$$\sum_{i,j}(a_j D_i^+ v, a_i D_j^+ v) = \Big\|\sum_i a_i D_i^- v\Big\|^2 - C[av] + O(k^2)\|v\|^2, \tag{7.14}$$

$$\Big\|\sum_i a_i D_i^+ v\Big\|^2 \le k\|D^- v\|^2 - C[av] + O(k^2)\|v\|^2, \tag{7.15}$$

$$\|D^- v\|^2 = \|D^+ v\|^2 + \int_Y \operatorname{tr} C\,|v|^2\, dV_\gamma. \tag{7.16}$$

Here and below a form denoted $O(k^2)\|v\|^2$ has absolute value at most a fixed constant times that quantity. To verify the first two formulas, move a D_j^+ across the pairing and commute $D_j^- D_i^+$ to $D_i^+ D_j^- - C_{ij}$. The reordered term in (7.15) is at most $k\|D^- v\|^2$, because $\sum_i |a_i w|^2 = k|w|^2$ for $w \in \mathcal{H}_k$. The extra commutator is curvature: it acts as a sum of rotations of norm $O(k)$ at degree k, and the two contractions supply one more factor k. This proves the claimed error bounds. Expanding the two squares proves (7.16). For complex functions all the displayed forms are understood through their real parts.

Substituting (7.14) into (7.13) rewrites that expression as

$$\begin{aligned}(2+2/d)\left[d\|D^+ v\|^2 - \Big\|\sum_i a_i D_i^+ v\Big\|^2 - dC[av]\right]\\ + 2\Big\|\sum_i a_i D_i^- v\Big\|^2 + O(k^3)\|v\|^2.\end{aligned} \tag{7.17}$$

We must control the remaining negative divergence square while retaining both an ordinary derivative estimate and an additional half-order angular derivative estimate. The latter requires a coefficient of size k/β in front of $|D^+v|^2$, including when k is much larger than β. Choose a small constant $\vartheta > 0$. On the fraction $1-\vartheta$ of its negative divergence square use (7.15)–(7.16). On the remaining fraction use the pointwise convex combination

$$\begin{aligned}\Big|\sum_i a_i D_i^+ v\Big|^2 &\le (1-\zeta)(k+n-1)|D^+v|^2 \\ &\quad + \zeta\left(2\Big|\sum_i a_i D_i^- v\Big|^2 + 8k^3\beta^2|v|^2\right), \qquad \zeta = c_1/\beta.\end{aligned} \tag{7.18}$$

The fraction $1-\vartheta$ in the integrated estimates is constant; the y-dependent ζ is used only in this pointwise inequality. The first bound follows from the adjoint multiplication operators, since the contraction $(q_i) \mapsto \sum_i a_i q_i$ has squared norm at most $k+n-1$ on polynomials of degree k. The second follows from $D_i^+ + D_i^- = 2kf_i$ and $|\sum_i f_i a_i v|^2 \le k\beta^2|v|^2$.

Combining these bounds in (7.17), the coefficient retained for $|D^+v|^2$ is

$$(2+2/d)\big[d-k-\vartheta(n-1)+\vartheta\zeta(k+n-1)\big]. \tag{7.19}$$

Since $d-k=(n-2)/2$, a sufficiently small fixed ϑ makes this at least a fixed positive constant plus a positive multiple of $c_1 k/\beta$. The coefficient of $|\sum_i a_i D_i^- v|^2$ is nonnegative when β_0 is sufficiently large. The Hessian contribution is exactly

$$-(2+2/d)\left[(d-1+\vartheta)C[av] + (1-\vartheta)k\int_Y \operatorname{tr} C\,|v|^2\,dV_\gamma\right]. \tag{7.20}$$

For $k \ge 1$ and $n \ge 3$, both coefficients inside this expression are positive. The Hessian hypothesis bounds it below by $ck^3\int\beta|v|^2$. This absorbs the curvature error $O(k^3)\|v\|^2$ by increasing β_0, and absorbs the last term of (7.18) by choosing c_1 sufficiently small. We have therefore retained positive multiples of

$$\|D^+v\|^2, \qquad \int_Y \frac{k}{\beta}|D^+v|^2\,dV_\gamma, \qquad k^3\int_Y \beta|v|^2\,dV_\gamma.$$

The first controls both $\|\nabla v\|^2$ and $k^2\int\beta^2|v|^2$, since its cross term is $-k\int\Delta_\gamma f\,|v|^2 \ge 0$. For the weighted derivative use the pointwise inequality

$$\frac{k}{\beta}|\nabla v|^2 \le 2\frac{k}{\beta}|D^+v|^2 + 2k^3\beta|v|^2.$$

No differentiation of β^{-1} is required. Dividing by N_k, and then summing the orthogonal degree components, proves (7.12); the weights $(1+k)^s$ are comparable to k^s because the zero degree is excluded. □

7.4 Keeping the principal perturbations

The raising estimate controls the model operator. The actual equation (7.5) also contains second base derivatives and a small second-order angular perturbation. We retain these terms in a comparison of squared norms.

Proposition 7.3 (Coercivity for the moving trace equation)**.** *Fix a base region with bounded reference geometry and the constants in* (7.1)*. Suppose P is the operator* (7.5)*, with the trace operators of Lemma 7.1. There exist β_0 sufficiently large, $\varepsilon_0 > 0$ and $C < \infty$, depending only on these fixed bounds, such that* (7.3) *with $\varepsilon \le \varepsilon_0$ implies*

$$\mathcal{N}(\varphi) \le C\|P\varphi\| \tag{7.21}$$

for every smooth, compactly base-supported function φ with zero mean on each sphere. These constants are independent of the shrinking parameter. Only the eight indicated derivatives of the small tensor h are required.

Proof. Let a dagger denote the full adjoint for $dV_\gamma d\sigma$. Covariant integration and the degree shift give

$$Z := T^\flat - T^\dagger = \sum_i f_i A_i. \tag{7.22}$$

A direct expansion yields

$$\begin{aligned} &\|(2T+U)\varphi\|^2 - \|(2T^\flat + U^\dagger)\varphi\|^2 \\ &\qquad = 4(\|T\varphi\|^2 - \|T^\flat\varphi\|^2) + \langle \mathcal{E}\varphi, \varphi\rangle, \end{aligned} \tag{7.23}$$

where

$$\mathcal{E} = 2[T^\dagger, U] + 2[U^\dagger, T] + [U^\dagger, U] - 2(UZ + (UZ)^\dagger). \tag{7.24}$$

We will prove

$$|\langle \mathcal{E}\varphi, \varphi\rangle| \le C\varepsilon \mathcal{N}(\varphi)^2. \tag{7.25}$$

Here are the complete order and weight counts needed for this claim. An operator has count (m, s) if its coefficient at ∂_y^α is angular order $m - |\alpha|$, of size $O(\beta^{s-|\alpha|})$, with the derivative allowances already specified. Adjoints preserve the count, and products add the counts. When a base derivative hits a coefficient it consumes one differential order and adds a factor β, so it preserves the weight index. A commutator of scalar angular operators loses one angular order: the products of their principal symbols cancel. Hence a scalar commutator loses one in total order while preserving the sum of the weight indices. The counts of $T, T^\dagger, U, Z$ are, respectively,

$$(2,1), \quad (2,1), \quad (2,0), \quad (1,1),$$

and every occurrence of U carries the factor ε. Thus (7.24) has count at most $(3,1)$ and a factor $O(\varepsilon)$; the quadratic U term is smaller.

There is a further cancellation that the total count alone does not express: no third base derivative survives. To check it explicitly, use an orthonormal sphere trivialization and write the base measure as $\mu(y)\,dy$. The sphere probability measure is then fixed. Write

$$U = a^{ij}\partial_i\partial_j + \mathcal{B}^i\partial_i + \mathcal{C}, \qquad a^{ij} = rQ^{ij} = a^{ji} \in \mathbb{R}.$$

The coefficients $\mathcal{B}^i$ and $\mathcal{C}$ are angular operators. Replacing a horizontal derivative by a coordinate derivative plus its angular connection term changes only base orders at most one, so these are all the second-base coefficients. If a star denotes the sphere adjoint, direct integration by parts gives

$$\begin{aligned} U^\dagger - U = &\left(2\mu^{-1}\partial_j(\mu a^{ij}) - \mathcal{B}^i - (\mathcal{B}^i)^*\right)\partial_i \\ &+ \mu^{-1}\partial_i\partial_j(\mu a^{ij}) - \mu^{-1}\partial_i(\mu(\mathcal{B}^i)^*) + \mathcal{C}^* - \mathcal{C}. \end{aligned}$$

In particular it has base order at most one. In $[T^\dagger, U]$ and its partner, the possible third-base-derivative coefficient is a commutator of an angular coefficient with rQ^{ij}, which is zero by (7.10). For $[U^\dagger, U]$, first write it as $[U^\dagger - U, U]$. The operator $U^\dagger - U$ has no second base derivative, since the coefficient in (7.10) is real. Taking adjoints against the smooth base density only produces lower base orders. Its remaining possible third-base-derivative coefficients commute for the same reason. Finally, UZ already has at most two base derivatives. It follows that $\mathcal{E}$ is a sum of terms of precisely the following three permitted types:

$$\varepsilon\beta^{-1}\Psi^1\partial_y^2, \qquad \varepsilon\Psi^2\partial_y, \qquad \varepsilon\beta\Psi^3. \tag{7.26}$$

This argument also covers lower orders, since $\beta \ge 1$.

For the first type in (7.26), integrate once in the base. The principal form is bounded by $C\varepsilon \int \beta^{-1}\|\partial_y\varphi\|_{1/2}^2$. A derivative of its coefficient costs one factor β and leaves an $\varepsilon\Psi^1\partial_y$ term. This includes differentiation of the weight itself: $|d\beta| \le C\beta^2$ implies $|d(\beta^{-1})| \le C$. The second type has the form bound

$$C\varepsilon \int_Y \|\partial_y\varphi\|_{1/2}\|\varphi\|_{3/2}\, dV_\gamma \le C\varepsilon \int_Y \left(\beta^{-1}\|\partial_y\varphi\|_{1/2}^2 + \beta\|\varphi\|_{3/2}^2\right) dV_\gamma.$$

The third type is bounded by $C\varepsilon \int \beta\|\varphi\|_{3/2}^2$. The lower-order term from differentiating a coefficient satisfies the same bound. Replacing coordinate derivatives by horizontal derivatives adds a bounded angular first derivative, again controlled by these norms. A fixed finite partition of the base region has the same harmless lower-order errors. This proves (7.25).

The constants in the preceding estimates are uniform for all $\beta \ge \beta_0$ once a fixed lower threshold is met. Choose ε_0 small relative to the constant in Lemma 7.2. Dropping the nonnegative second squared norm in (7.23) gives $\mathcal{N}(\varphi) \le C\|(2T+U)\varphi\|$. Finally,

$$\|J\varphi\| \le C\|\varphi\|_{L^2(Y;H^1(S_y))} \le C\beta_0^{-1}\mathcal{N}(\varphi),$$

so a sufficiently large β_0 absorbs J and proves (7.21).

All these operations use angular symbol estimates pointwise in the base, and ordinary base differential operators of degree at most two. Formal adjoints, the commutator expansion and the one integration by parts use at most the five base coefficient derivatives specified in (7.9). Construction of these coefficients differentiates the moving trace operators once and the second metric to only finitely many orders below eight. Higher angular symbol seminorms depend on the fixed smooth first-metric family; differentiating in the angular variable does not differentiate the center-dependent tensor h. Thus the eight smallness derivatives in (7.3) suffice in every fixed dimension. No smallness assumption on all derivatives of h is used. □

7.5 The constant mode and the uniform density bound

We now apply the coercive estimate to the transfer density constructed in Section 6, restoring the notation $r = r_\delta$. Its normalization is $\int_{S_y} \psi\, d\sigma_y = 1$, so $\chi(\psi - 1)$ has zero fiber mean for the fixed cutoff (6.9). This is the mean-zero condition required by Proposition 7.3.

The decomposition (7.8) gives a uniform bound for $P1$: its raising part annihilates 1, only U's bounded zero-base-derivative coefficient acts on 1, and J is uniformly bounded in Ψ^1. There is no factor growing with β in this assertion.

For any fixed smooth base cutoff, the same decomposition gives

$$\|[P,\chi]w\| \le C_\chi \left(\|w\|_{L^2(\operatorname{supp} d\chi;H^1(S_y))} + \|r\nabla w\|_{L^2(\operatorname{supp} d\chi\times S_y)}\right), \tag{7.27}$$

where the support can be replaced by a fixed small neighborhood of the supports of all derivatives of χ. Indeed $[2T,\chi] = 2\sum_i A_i^*(\nabla_i\chi)$, the second base derivative of U contributes $2rQ^{ij}(\partial_i\chi)\partial_j$ and a bounded zeroth-base-derivative term, and its first base derivative contributes a bounded angular order-one term. The operator J commutes with χ.

Lemma 7.4 (Uniform transfer density)**.** *Fix the structural parameters so that Propositions 6.2 and 7.3 apply. There is a constant C independent of δ and of sufficiently small spatial dilations λ such that*

$$\|\chi\psi\|_{L^2(S_\gamma Y)} \le C \tag{7.28}$$

whenever the coefficient condition (6.10) holds.

Proof. Set $\varphi = \chi(\psi - 1)$, which is compactly supported in the base and has zero fiber mean. For the actual moving-sphere operator P, Equation (7.5) gives

$$P\varphi = P(\chi(\psi-1)) = -\chi P1 + [P,\chi](\psi - 1). \tag{7.29}$$

The cutoff estimate (7.27) involves only angular derivatives through order one and $r\nabla\psi$ on the supports of derivatives of χ. Proposition 6.3 bounds these there uniformly: on the lower region the density is that of ordinary evaluation, and on the other region the radii stay at a fixed positive separation. Together with the bound for $P1$, this gives a uniformly bounded L^2 norm for the right side of (7.29). Proposition 7.3 bounds φ; adding the fixed function χ proves (7.28). □

The uniform bound now applies to the density representing the actual Green functional at every radius allowed by (6.10). Exact affine moments will turn it into a two-power bound for the paired harmonic differences.

8 From harmonic differences to local metric extension

Lemma 7.4 bounds the transfer density uniformly on the shrinking spheres. Exact reproduction of affine functions now turns that bound into a two-power estimate for the difference of paired harmonic functions. The harmonic Hessians convert derivatives of these differences into metric derivatives. Finite-order interpolation will then improve the temporary metric bound and allow the radii to shrink with the spatial dilation held fixed.

8.1 From the functional to harmonic jets

Return to the finite harmonic family in Theorem 5.1. Its values and first derivatives at 0 agree between the two metrics, because the functions agree on the one-sided open set and are smooth. After dilation define

$$v^a_{j,\lambda}(y) = \lambda^{-2}(u^a_j(\lambda y) - u^a_1(0) - \lambda\, du^a_1(0)\cdot y). \tag{8.1}$$

The subtracted functions are harmonic because the coordinates are harmonic. These new pairs still agree on the known side and are transferred by T_y. For each fixed integer m they have uniform C^m bounds on every required fixed bounded coordinate neighborhood. For derivative orders $j \geq 2$, this follows from the factor λ^{j-2}; orders zero and one follow from the second-order Taylor remainder. Moreover,

$$\partial^2 v^a_{1,\lambda}(y) \longrightarrow \partial^2 u^a_1(0)$$

smoothly on fixed sets. Thus the Hessian spanning condition has a uniform rank bound after sufficiently small dilation. Write $V^a = v^a_{2,\lambda} - v^a_{1,\lambda}$.

Lemma 8.1 (Two-power moment bound)**.** *Under the coefficient condition (6.10), the paired differences satisfy*

$$\left\| \frac{\chi V^a}{r_\delta^2} \right\|_{L^2(Y)} \leq C \tag{8.2}$$

with a constant independent of δ and sufficiently small λ.

Proof. Fix a center y and subtract from $v^a_{1,\lambda}$ its affine coordinate Taylor polynomial at y. Exact constant and coordinate reproduction in (6.15) cancels this polynomial in $T_y v^a_{1,\lambda} - v^a_{1,\lambda}(y) = V^a(y)$. The remainder on $\partial B_\gamma(y,r)$ is bounded by Cr^2, since coordinate distance and γ-distance are uniformly comparable and the functions have uniform second derivatives. Hence

$$|V^a(y)| \leq C r_\delta(y)^2 \|\psi(y,\cdot)\|_{L^2(S_y)}.$$

Lemma 7.4 proves (8.2). The coordinate volume measure and dV_γ are uniformly comparable. No derivative in y of the continued density is used here. □

We next express metric derivatives in terms of derivatives of the paired differences. Recovery of the conformal metric class from the hyperplane of harmonic Hessians is also used in [4, Proposition 4.1 and Remark 4.2]. All Hessians in this step are ordinary coordinate Hessians, not covariant Hessians. Because the coordinates are harmonic, the equations for $v^a_{j,\lambda}$ have no first-order term in these coordinates:

$$(g^{-1}_{k,\lambda})^{ij}\partial_{ij}v^a_{k,\lambda} = 0 \qquad (k = 1, 2).$$

Multiplication by the scalar factors in (6.2) therefore gives

$$h^{ij}\partial_{ij}v^a_{1,\lambda} = -Q^{ij}\partial_{ij}V^a, \qquad \operatorname{tr}(\gamma h) = 0. \tag{8.3}$$

Harmonicity puts each $\partial^2 v^a_{1,\lambda}(y)$ in the hyperplane annihilated by $\gamma^{-1}(y)$. The uniform rank bound established above shows that these matrices span this hyperplane for small λ. The extra equation $\operatorname{tr}(\gamma h) = 0$ completes an injective system on symmetric contravariant matrices: an element annihilating all those Hessians is a multiple of γ^{-1}, and its γ-trace is n times that multiple. The associated finite matrix has a smooth left inverse with uniform derivatives on Y. For example, in fixed coordinate bases use $(A^*A)^{-1}A^*$, where A is this injective matrix. Its smallest singular value stays bounded below by the spanning condition and smooth convergence under dilation. Differentiating (8.3) at most eight times gives

$$H_8(y) \le C \sum_a \sum_{|\alpha|\le 10} |\partial^\alpha V^a(y)|. \tag{8.4}$$

The derivatives of Q, γ, and of the finite harmonic family are uniformly bounded. In particular, this inversion introduces no factor involving r_δ^{-1}.

The metric estimate requires derivatives of the paired differences through order ten, whereas Lemma 8.1 controls only their weighted L^2 norm. The following finite-order interpolation estimate bridges these two statements using the already fixed smooth bounds.

Lemma 8.2 (Finite-order interpolation)**.** *Let V be smooth on a neighborhood of $\overline{B(y,s)} \subset \mathbb{R}^n$, with $0 < s \le 1$ and $\|V\|_{C^m} \le A_m$ there. For every multi-index α of length $j < m$,*

$$|\partial^\alpha V(y)| \le C_{m,n}\big(s^{-j-n/2}\|V\|_{L^2(B(y,s))} + A_m s^{m-j}\big). \tag{8.5}$$

Proof. Expand $V(y + sz)$ at $z = 0$ through degree $m - 1$ on the unit ball. The remainder has absolute value at most $C_{m,n}A_m s^m$. The L^2 norm of the Taylor polynomial is therefore at most $s^{-n/2}\|V\|_{L^2(B(y,s))} + C_{m,n}A_m s^m$. On the finite-dimensional space of polynomials of degree at most $m - 1$, every coefficient is bounded by a constant times the L^2 norm on the unit ball. Its z^α coefficient is $s^j\partial^\alpha V(y)/\alpha!$, giving the claim. □

8.2 A strict improvement of the metric bound

Set

$$D = 10 + n/2, \qquad \eta = \frac{1}{2D}, \qquad p > 2D, \qquad m \ge \lceil 10 + 3D \rceil. \tag{8.6}$$

We also take the integer p large enough for Proposition 7.3. Thus $\eta > 1/p$, and m is one fixed finite smoothness order. The constants may depend on these choices, which precede the final spatial dilation and all shrinkage.

Lemma 8.3 (Superlinear improvement)**.** *After the choices in (8.6), there are $C > 0$ and $\rho_0 > 0$, independent of δ and sufficiently small λ, such that the coefficient condition (6.10) implies*

$$H_8(y) \le C r_\delta(y)^{3/2} \quad \text{for } y \in K \text{ with } r_\delta(y) < \rho_0. \tag{8.7}$$

Proof. Put $r = r_\delta(y)$ and $s = r^\eta$. By Lemma 6.1, $r_\delta^{1/p}$ has a fixed Lipschitz constant. Since

$$\frac{s}{r^{1/p}} = r^{\eta - 1/p} \longrightarrow 0,$$

a sufficiently small threshold ρ_0 ensures

$$\tfrac{1}{2} r^{1/p} \le r_\delta(z)^{1/p} \le \tfrac{3}{2} r^{1/p} \quad (z \in B(y,s)).$$

The same fixed threshold ensures that this ball lies where $\chi = 1$, because K has a fixed buffer inside that set. The upper comparison and Lemma 8.1 imply

$$\|V^a\|_{L^2(B(y,s))} \le C r^2.$$

Apply Lemma 8.2 and the fixed uniform C^m bounds of the renormalized functions. For $j \le 10$ it gives

$$|\partial^\alpha V^a(y)| \le C_m \big(r^{2-\eta(j+n/2)} + r^{\eta(m-j)} \big) \le C_m r^{3/2}.$$

Both exponents are at least $3/2$ by (8.6). Equation (8.4) completes the proof. □

The gain in (8.7) is stronger than the temporary bound $H_8 \le \varepsilon r_\delta$. Its usefulness is that a single choice of the spatial dilation makes the improvement strict at every radius, including those above the small threshold.

8.3 Local extension of the isometry

Proof of Theorem 5.1. Make the linear normalization and spatial dilation of Section 6. All harmonic functions and their finite spanning family are chosen before the dilation. We now specify the remaining choices and hold each one fixed before making the next.

First choose the fixed conformal geometry, cylinder, depth function, and cutoff, keeping uniform background smooth bounds for $0 < \lambda \le \lambda_0$. Choose p, η, m as in (8.6), with p large enough for the geometric and angular estimates. Choose ε for those estimates and then R_* so that the balls lie in normal neighborhoods and $r_\delta \beta_\delta^2 \le \varepsilon$ for every δ. These choices fix every separation range needed to start the continuation and to bound the cutoff terms. Restrict λ_0 further so that Proposition 5.5 applies to these fixed ranges, with bounds uniform for $0 < \lambda \le \lambda_0$.

The constants C, ρ_0 of Lemma 8.3 are now fixed. Choose

$$0 < \rho < \rho_0, \qquad C\rho^{1/2} \le \varepsilon/2. \tag{8.8}$$

At points with $r_\delta < \rho$, that lemma improves the coefficient bound to $H_8 \le \varepsilon r_\delta/2$ whenever the temporary bound holds. At points with $r_\delta \ge \rho$, the same improvement will hold if

$$\sup_Y H_8 \le \varepsilon\rho/2. \tag{8.9}$$

Since $h \to 0$ smoothly under dilation, choose one $\lambda \le \lambda_0$ satisfying (8.9) and the additional initial bounds described next. This λ is kept fixed for the rest of the proof.

There is a uniform reason the coefficient bound holds off K for every δ. Outside K, either $y_n < -1/2$, where $h = 0$ after the fixed dilation is sufficiently small, or $t > 2t_b$, where

$$r_\delta \ge R_*(2t_b)^p > 0.$$

Smooth smallness of h therefore gives $H_8 \le \varepsilon r_\delta/2$ off K for every δ by reducing the same dilation once. At $\delta = 1$, the radius has a positive minimum on all of Y, so we can also ensure the initial coefficient bound there. These conditions involve only fixed positive constants, and are compatible with (8.9).

For this fixed metric pair on the dilated neighborhood, set

$$\mathcal{A} = \{\delta \in (0,1] : H_8(y) \le \varepsilon r_\delta(y) \text{ for every } y \in K\}.$$

It contains 1. At any $\delta \in \mathcal{A}$, the bound already established off K gives the full hypothesis (6.10). Proposition 6.2 and the uniform density estimate of Lemma 7.4 then apply. The moment bound, metric system, and interpolation argument give Lemma 8.3. The small-radius improvement (8.8) and the large-radius bound (8.9) give

$$H_8 \le \tfrac{1}{2}\varepsilon r_\delta \quad \text{on } K.$$

For every fixed positive δ, r_δ has a positive minimum on the compact set K. Thus $\delta \mapsto \max_K(H_8/r_\delta)$ is continuous on $(0,1]$. The set $\mathcal{A}$ is relatively closed, and the strict improvement makes it relatively open. Connectedness implies $\mathcal{A} = (0,1]$.

Finally let $\delta \downarrow 0$, with the spatial dilation, harmonic family, cutoff and all structural parameters fixed. On the fixed open neighborhood of 0 where $t < 0$, the radii tend to zero. Hence $h = 0$ there. By (6.2), $g_{2,\lambda}$ is conformal to $g_{1,\lambda}$ on that neighborhood. Write $g_{2,\lambda} = e^{2\omega} g_{1,\lambda}$. The conformal change formula for the Laplace–Beltrami operator is

$$\Delta_{e^{2\omega}g} u = e^{-2\omega}(\Delta_g u + (n-2)\langle d\omega, du\rangle_g).$$

Apply it to every common harmonic coordinate. Since their differentials form a basis and $n - 2 \neq 0$, it forces $d\omega = 0$. Since $g_{1,\lambda}(0) = g_{2,\lambda}(0) = I$, this constant is zero. This direct conformal calculation also appears in [4, Proposition 4.1]. The metrics therefore agree on a connected neighborhood of 0; undoing the dilation proves the theorem. □

For clarity, the two limits in this proof have different roles: spatial dilation is used once to obtain a fixed small metric difference, and the subsequent shrinking family improves the bound for that same difference. The parameter order is

$$\begin{aligned} \text{fixed geometry and harmonic family} &\longrightarrow (p, \eta, m) \longrightarrow (\varepsilon, R_*) \\ &\longrightarrow \rho \longrightarrow \lambda \text{ fixed} \longrightarrow \delta \downarrow 0. \end{aligned} \tag{8.10}$$

All estimates use finitely many derivatives selected before either limiting step. Theorem 5.1 is the local input that will prevent a maximal matched region from having an interior frontier.

9 The global isometry and the measured boundary

We now apply Theorem 5.1 to the data constructed in Section 4. The source evaluations identify which points should correspond. Their separation makes this correspondence single-valued, and their behavior at the boundary prevents its escape from the interior. The local theorem then removes every interior frontier of the isometric correspondence. The use of a maximal matching set and limits of source evaluations follows the source-embedding architecture in [4, Section 6.1]; Theorem 5.1 supplies the smooth continuation step needed here.

Retain the extended manifolds M_j^e, their common cap E, the open source set $U_0 \Subset E$, and the functionals I_j of Equation (4.4). For this section an *admissible match* is a smooth local isometry

$$\Phi : W \longrightarrow (M_1^e)^\circ, \qquad W \subset (M_2^e)^\circ,$$

where W is connected, open, and contains E, such that

$$\Phi|_E = \mathrm{Id}, \qquad I_2(p) = I_1(\Phi(p)) \quad (p \in W). \tag{9.1}$$

Thus $\Phi^* g_1 = g_2$ on W. The identity on E is an admissible match by Proposition 4.2.

Proposition 9.1 (No interior frontier)**.** *There is an isometry*

$$\Phi : (M_2^e)^\circ \longrightarrow (M_1^e)^\circ$$

onto the first interior satisfying Equation (9.1)*.*

Proof. The maximal match and its Green kernel. Two admissible matches agree wherever their domains overlap, since I_1 separates interior points by Lemma 4.3. Their union therefore defines a single smooth local isometry on the union W of all admissible domains. This union is connected because every domain contains the connected set E. The union map is injective: equal images give equal I_2 evaluations, hence equal original points. It is itself an admissible match and is maximal by construction.

We first record the Green equality on this larger region:

$$G_2(p,q) = G_1(\Phi(p),\Phi(q)), \qquad p,q \in W, \quad p \neq q. \tag{9.2}$$

For fixed $p \in W$, equality of its evaluations and the common source density give this equality for the other variable in U_0, initially in the distributional sense. Both Green functions are smooth away from their pole. After pullback by the local isometry, their difference is harmonic on $W \setminus \{p\}$. This set is connected, and equality holds on a nonempty open part of it. Unique continuation proves Equation (9.2).

A frontier partner and common coordinates. Suppose W is not the entire second interior. Connectedness gives an interior frontier point p. If $p_k \in W$ tends to p, compactness of M_1^e gives a subsequential limit q of $\Phi(p_k)$. For every fixed source f, continuity up to the boundary gives

$$I_2(p)(f) = I_1(q)(f).$$

The point q is interior: all boundary evaluations are zero, whereas $I_2(p)$ is nonzero. Any two such limits have the same evaluations and so coincide. It follows that every matched sequence tending to p has partners tending to this unique q. In particular, nearby matched points have partners in any prescribed neighborhood of q.

Both p and q are outside $\overline{U_0}$. The first statement holds because $\overline{U_0} \subset E \subset W$. If q belonged to $\overline{U_0}$, the common-cap identification would give $I_1(q) = I_2(q)$; separation on the second manifold would force $p = q \in E$, a contradiction.

Choose sources $f_1, \dots, f_n$ so that

$$y = (u^2_{f_1}, \dots, u^2_{f_n})$$

has independent differentials at p, and put $x = (u^1_{f_1}, \dots, u^1_{f_n})$ near q. The second tuple is also a coordinate system. Indeed the isometries at the points p_k give equality of the gradient Gram matrices

$$\big(\langle du^2_{f_a}, du^2_{f_b}\rangle_{g_2}(p_k)\big)_{a,b} = \big(\langle du^1_{f_a}, du^1_{f_b}\rangle_{g_1}(\Phi(p_k))\big)_{a,b}.$$

The first limit is positive definite, and so is the second. The coordinate values at p, q coincide. Restricting the charts gives a common coordinate ball V about this value, with closures away from $\overline{U_0}$. The preceding convergence observation ensures that every already matched point sufficiently near p has its partner in the first chart. In these coordinates its identification is $x = y$.

Let $A \subset V$ be the coordinate image of the already matched second-side points. It is open and has the center value in its relative frontier. The two coordinate metrics and every paired source function agree on A. By Lemma 4.3, choose finitely many first-side source functions whose ordinary Hessians span the hyperplane annihilated by g_1^{-1} at the center. After shrinking V, their spanning rank has a positive lower bound throughout V. Their second-side partners are harmonic there and agree with them on A.

A touching ball removes the frontier. We can now find a smooth ball touching the unknown part, without assuming any regularity of the frontier of A. Choose $c \in A$ sufficiently close to the center that

$$r = \operatorname{dist}(c, V \setminus A) > 0, \qquad \overline{B(c,r)} \Subset V.$$

For example, if the center has distance R from ∂V, take c within $R/3$ of it. Then $r < R/3$, and a minimizing sequence stays in a compact subset of V. Thus the distance is attained at some

$$z_* \in \partial B(c,r) \cap (V \setminus A).$$

On $B(c,r)$ the metrics, the finite harmonic pairs, and the Green kernels in both variables agree. Smoothness gives equality of the metric and harmonic-function jets at z_*, and the Hessian spanning condition holds there by our choice of V. All the inputs of Theorem 5.1 are therefore satisfied at z_*. It gives equality of the coordinate metrics on a neighborhood of z_*. We may take a small connected ball there meeting $B(c,r)$. Each paired source difference is now harmonic for the common metric and vanishes on that intersection. Unique continuation makes every pair agree on the same ball. The resulting local isometry thus matches all evaluations.

This isometry agrees with the existing one on every overlap, by separation of evaluations. Its union with W is an admissible match whose domain contains z_*, contradicting maximality. Consequently $W = (M_2^e)^\circ$.

Surjectivity. The image is open because Φ is a local diffeomorphism. It is also closed in the first interior. Indeed, if $\Phi(p_k)$ tends to an interior point q, a subsequence of p_k converges in the compact second manifold to p. Evaluation continuity again gives $I_2(p) = I_1(q)$, so p cannot lie on the boundary. Then p is in the domain and $\Phi(p) = q$. Connectedness of the first interior makes the image all of it. An injective surjective local diffeomorphism has a smooth inverse, proving the proposition. □

9.1 Returning to the original manifold

The interior isometry still has two duties: it must extend smoothly through the original boundary, and it must fix every point of the measured patch, including components where no cap was attached.

Proposition 9.2 (Smooth completion and the accessible gauge)**.** *The isometry of Proposition 9.1 extends to a smooth diffeomorphism $M_2^e \to M_1^e$. It preserves the original copies of M, and its restriction satisfies*

$$g_2 = \Phi^* g_1, \qquad \Phi|_\Gamma = \mathrm{Id}.$$

Proof. An interior isometry preserves lengths of curves and hence the intrinsic distances. The metric completion of the interior of a compact smooth Riemannian manifold with boundary is the manifold itself with its length metric. To see the only boundary issue, use a collar to push a boundary-touching curve an arbitrarily small distance into the interior; its length changes arbitrarily little. Compactness and local metric comparison then identify the completion with the original compact manifold. Thus Φ and its inverse extend to mutually inverse homeomorphisms of the two completions. Boundary is mapped to boundary, since the interiors already correspond bijectively.

These extensions are smooth. In a sufficiently small uniform boundary collar, the distance s to the boundary is a smooth function, and its unit gradient is the inward normal flow. The isometry preserves s and, in the interior, intertwines its gradients. Fix a small positive level $s = s_0$. The map is already smooth on that interior hypersurface. Expressing points with $0 \le s \le s_0$ by normal flow from this level expresses Φ as the composition of this smooth level map with smooth normal flows on both sides. This extends smoothly to $s = 0$. The same argument applies to the inverse.

Since Φ is the identity on E, continuity makes it the identity on $\overline{E}$. The original interiors satisfy

$$M_j^\circ = (M_j^e)^\circ \setminus \overline{E}.$$

The extended map preserves these complements, and hence their closures, the original manifolds. Its restriction is therefore a smooth diffeomorphism $\Phi : M \to M$ with $g_2 = \Phi^* g_1$. It fixes the attached original patch $P_0 = \{b > 0\}$ pointwise.

To prove the full boundary assertion, take any $f \in C_c^\infty(\Gamma)$ and let v_f^j now denote the harmonic functions on the *original* manifolds with boundary value f. On the second manifold both v_f^2 and $v_f^1 \circ \Phi$ are g_2-harmonic. Their difference has zero boundary value on P_0, where Φ is the identity. It also has zero normal derivative there. In fact the equal energy maps give

$$(\partial_{\nu_1} v_f^1)\, dS_{g_1} = (\partial_{\nu_2} v_f^2)\, dS_{g_2} \quad \text{on } P_0.$$

The boundary densities are the same by Lemma 4.1, and the smooth isometry transports the outward unit normal. Boundary unique continuation, followed by interior continuation, therefore gives

$$v_f^2 = v_f^1 \circ \Phi \quad \text{throughout } M.$$

Taking boundary traces yields

$$f(q) = f(\Phi(q)) \qquad (q \in \partial M,\ f \in C_c^\infty(\Gamma)).$$

For $q \in \Gamma$, these test functions distinguish q from every other boundary point: choose a function equal to one at q with support in a small neighborhood avoiding the other point. Thus $\Phi(q) = q$ for all $q \in \Gamma$. □

Proof of Theorem 2.2. Lemma 4.1, Proposition 4.2, and Lemma 4.3 provide the common cap, Green kernels, and finite harmonic tests. Theorem 5.1 applies at every interior contact in Proposition 9.1. Proposition 9.2 gives the required diffeomorphism. All integrations use densities; the argument required neither an orientation nor connectedness of the boundary or of Γ. □

10 The full ordinary Dirichlet-to-Neumann map

Proof of Theorem 2.1. The common tangential metric gives $dS_{g_1} = dS_{g_2} = dS$. For smooth boundary functions Green's formula gives

$$Q_{g_j}(f, h) = \int_{\partial M} (\Lambda_{g_j} f)\bar{h}\, dS.$$

Equality of the ordinary maps therefore gives equality of the energy forms, first on smooth functions and then on $H^{1/2}$ by their boundedness. Choose a boundary point p and set $\Gamma = \partial M \setminus \{p\}$. Since $\dim \partial M = n - 1 \geq 2$, this is nonempty, proper, relatively open, and dense, including when the boundary is disconnected. Theorem 2.2 supplies a smooth Φ with $g_2 = \Phi^* g_1$ fixing Γ. Continuity fixes the remaining point. Taking $F = \Phi^{-1}$ gives $g_1 = F^* g_2$ and fixes every boundary point, as required. □

The data used are exactly one full, ordinary, zero-frequency map. All dimensions $n \geq 3$, arbitrary smooth positive metrics, nonorientable manifolds, and arbitrary boundary components have been retained.

AI use. This exposition was prepared with OpenAI Codex (GPT-6.1 Sol), which assisted with source retrieval, proof reconstruction, mathematical checks, typesetting, and revision. The underlying uniqueness argument is the OpenAI preprint [5], whose attribution is retained. No formal verification or independent human refereeing is claimed.

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